Linköping University | Institute for Analytical Sociology
Master's thesis, 30 ECTS | Computational Social Science
2026 |

Activating Antagonism
– Populism and Ego-Network Structure among German MPs on Twitter

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Aktiverande Antagonism
– Populism och egonätverksstruktur bland tyska förbundsdagsledamöter på Twitter

Thomas Haase

Supervisor: Károly Takács
External supervisor: Philipp Lorenz-Spreen
Examiner: Miriam Hurtado Bodell

LINKÖPING UNIVERSITY

Linköping University
SE–581 83 Linköping
+46 13 28 10 00, www.liu.se

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Abstract

The thesis explores how populist rhetoric by German parliamentary members shapes the structure of engagement communities on Twitter. From a minimal definition of populism, a systems-theoretic, social-psychology-informed account of populism's meso-level dynamics is developed. This extends prior work centred on engagement metrics toward the structural question of whether politicians usage of populism on Twitter induces denser local engagement networks around the politician. The analysis is based on a dataset of German MPs' Twitter posts from 7.–14. February 2022 combined with the reply threads beneath the tweets. A Qwen3-235B LLM rates tweets on core dimensions of populism through few-shot classification with an expert-validated system prompt. The rated tweets are combined into a user-level reply network. Politicians' ego networks are extracted and filtered to regress a politician's use of populism on the connectedness of the respective engagement community. Populist German MPs occupy more strongly connected engagement communities than non-populist MPs. The impact of alters on tie creation is additionally explored. Alter interaction patterns show slight populism-homophily, with populist alters' replies bridging alters of differing activity volumes. Populist community formation is consistent with established self-selection accounts of populist mobilisation.

Acknowledgements

Luhmann called sociology “Die Abklärung der Aufklärung” in his essays *Sociological Enlightenment I*. He drew a distinction between the individual-focused enlightenment and the sociological mission of focusing on an individual’s contextual dependencies. In the same way, my improbable path to Sweden was shaped by all the amazing people around me. I deeply thank all of you from the bottom of my heart.

First, I thank my supervisors and mentors Károly Takács and Philipp Lorenz-Spreen for the kind guidance, all the inspiring meetings and, most importantly, the priceless support on my academic journey.

I especially want to thank all the dear and fantastic friends I made in Norrköping. I will miss all the fun evenings, the hours we spent in the library and most importantly all of you. You helped me grow tremendously, and I am deeply thankful for that.

I also want to thank the research group at SynoSys for integrating me so effortlessly. The dinner situations and coffee talks shaped the way I think about methods and science significantly. It is hard to imagine a better place to write this thesis.

Although this is my master’s thesis, I want to extend my deep thanks to Elmar Schlüter for the outstanding support I received during my bachelor’s. You are a fantastic researcher and a great role model for me. Whenever I learn something new now, I realize that you laid the foundation for it years ago. I also want to thank all my dear friends in Gießen for the coffee chats in the library, lunches and dinners. It was a wonderful time.

Zuletzt und am Allerwichtigsten:
Danke Mama und danke Papa! Ihr seid die besten Eltern, die ich mir je hätte wünschen können. Ich habe euch sehr lieb.

Generative AI Statement

In accordance with the IAS generative AI policy, this statement documents the use of generative AI tools in the preparation of this thesis.

A distinction must be drawn between two categories of AI use. First, large language models are employed as a research instrument within the methodology itself, as described in the methods chapter. This methodological use forms part of the empirical contribution of the thesis and is documented and validated there in detail.

Second, generative AI was used in a limited capacity as an aid for complementary, non-core tasks, in line with what the policy explicitly permits. Such tools were occasionally consulted to support coding work, debugging scripts and resolving syntax or tooling issues. All code was reviewed, tested, and adapted by the author, and the analytical decisions remain the author's own.

Generative AI was not used to produce the substantive text of this thesis. The framing, argumentation, theoretical engagement, methodological reasoning, and writing are entirely the author's own work.

Ethics Statement

The data underlying this thesis was collected in 2022 by Dr. Armin Pournaki through the official Twitter API for academic researchers, such that no scraping or circumvention of platform restrictions was necessary and the platform itself was not adversely affected by the collection. All raw and processed data is stored locally and, where backups are required, on TU Dresden's secure Datashare; no commercial cloud service was used at any stage, and the LLM-based text analysis was conducted on TU Dresden's Capella HPC cluster using a locally hosted model, ensuring that the data never left the controlled research environments of MPI Leipzig, TU Dresden, and the author's local machine.

Ordinary users appear in the analysis only as constituent elements of aggregate engagement structures; no individual usernames or user-level identifiers from the general public are reported or otherwise made traceable.

Politicians, by contrast, are referred to by name, since their posting behaviour forms part of a public political discourse whose mechanics are of legitimate interest to the public they represent. This interest is not only morally defensible but also legally anchored in Article 40 of the Digital Services Act, which establishes platform-mediated political communication as a legitimate object of academic research on systemic risks (European Parliament and Council 2022).

1. Introduction

Social media threatens democracy through several channels, one of them being populism (Lorenz-Spreen et al. 2022). While scholars debate definitions, populist movements are democratizing neoliberalizing regimes (Brown 2020) and destabilizing Western democracies (Lorenz-Spreen et al. 2022). This is hardly surprising considering the essence of populism.

Populism research has historically been divided by geographical focus, methods, and host ideologies (Hunger and Paxton 2022). Since 2016 the research landscape has been dominated by political science perspectives converging on populism as a discursive practice (Tuğal 2021). They almost uniformly adopt Mudde (2004)'s definition of populism as "an ideology that considers society to be ultimately separated into two homogeneous and antagonistic groups, 'the pure people' versus 'the corrupt elite', and which argues that politics should be an expression of the volonté générale (general will) of the people" (Mudde 2004:542).

While theoretical work refines the concept largely within Marxist traditions, relational and systems-sociological perspectives have remained peripheral (Hunger and Paxton 2022; Tuğal 2021). This is puzzling, because Mudde's definition is itself relational: populism is constituted through the antagonism between two groups. Simmel already connected antagonism to group cohesion, noting that groups are "…held together by a shared aversion […] to which entirely foreign elements are drawn by the commonality of an antagonism." (Simmel 1908:Ch.4). A relational reading of Mudde is therefore not a far fetched but literally a theoretical operationalization of it.

One of computational social science's original aims is to make use of digital data for social science questions (Lazer et al. 2009). Since the dynamics of complex systems are baked into the design of social media, researchers should take a complex-systems perspective when studying them (Bak-Coleman et al. 2025). Yet existing large-scale observational social media studies of populism analyze only textual content (Erhard et al. 2025; Serrano, Papakyriakopoulos, and Hegelich 2020; Yarchi, Baden, and Kligler-Vilenchik 2021). Measured populism has rarely been researched in terms of structural effects.

This study fills that gap by asking the central question:

"How does populist rhetoric by German parliamentary members shape the structure of their reply communities on Twitter?"

Populism research is divided over whether populism is a thin or a thick ideology. The minimal definition reads it as a vertical antagonism between a homogeneous people and a homogeneous elite (Mudde 2004), often enriched with a horizontal dimension constructing the people additionally against outside groups (Brubaker 2017). While the thick ideology is threatening diversity and democratic systems the thin ideology is only antidemocratic when the people are homogenous and aim at replacing an diversity-preserving constitution (Abts and Rummens 2007; Yates and Mondon 2026). Social media is a low affordance medium and not a neutral arena due to recommender architecture actively steering who encounters whom (X 2026). Parties are movements competing for political power in this attention environment (Simon 1971; Tilly 1978:117), and populist rhetoric constructs the distinction of the pure people against a corrupt elite. This gives encounters a distinction around which they can stabilize through users selfselecting actively into conversations. Engagement in this medium is not agreement (Morselli et al. 2026) but a structural fact, users repeatedly showing up in the same place around the same politician whatever their reason for doing so, and whether such stabilization actually occurs is then a question about communication and not about speakers.

The methodological approach translates the outlined theoretical ideas by using a Twitter reply network to capture social interactions on social media. Compared to retweets and mentions, replies are a way of directly answering to a message another person published. The dataset contains politicians tweets and their replies to construct such a replynetwork by linking users through replies as a weighted directed network. To explore the effect of a politician using populist rhetoric on their community all tweets are labeled by the amount of populist attitude they communicate. This textanalysis is achieved through a fewshot LLM Qwen3-235B annotation using an expert validated systemprompt. The egonetworks of politicians are extracted and their structure compared by their usage of populist rhetoric to explore if the expected relationship can be observed. The results-section first describes the validation of the prompt and the distribution of populism dimensions across users, politicians and parties. After that the structure of the reply network is described. In the end the local interactions of politicians are captured through egonetworks. The egonetworks of politicians serve as the basis for linear regression, comparing the connectedness of users in the egonetwork around the populist and non-populist politicians. Extending this analysis the impact of alters on tie formation in the egonetworks is briefly adressed. The discussion then answers the researchquestion by framing the findings in terms of the chosen theoretical approach. In the end a final conclusion is drawn.

2. Literature Review

Populism research is divided over one main question: Is populism a thin or thick ideology? And if it is thick, how so? The following section provides an overview of the conflict and explains the implicit theoretical source of the conflict. The second half proposes a new generalizing perspective on the topic motivated through already observed micro mechanisms.

2.1. Populist Theories of Populism

2004 Cas Mudde proposed a minimal definition of populism as an ideology where a message is framed in a way that the powerless homogenous people stand against a ruling homogenous elite as antagonistic groups (Mudde 2004). This definition has been considered “thin”, because it only considers a vertical divide of society (Brubaker 2017). Vertical in the sense that one group is ruling the other. When read as an ideology Mudde (2004) s minimal populism implicitly includes a class component as the core of its definition (Yates and Mondon 2026). The definition is currently the most cited work on populism. This can be explained with the minimalism increasing the potential of connected communication for theorists discussing the work (Brubaker 2017; Luhmann 1987; Tuğal 2021; Urbinati 2019). Empirical studies operationalized the main dimensions of people-centrism and anti-elitism in surveys and finetuned bert classifier PopBert on german Bundestagspeeches (Akkerman, Mudde, and Zaslove 2014; Castanho Silva et al. 2020; Erhard et al. 2025). Often this minimal definition is combined with some other dimension like antipluralism or the distinction between left and rightwing populism (Castanho Silva et al. 2020; Erhard et al. 2025). Theorists combine the thin definition with institutional and substantive analyses when discussing populism but those are chosen rather arbitrary (Tuğal 2021). The minimal definition is part of the same family of subjectivist definitions as discoursive and performative marxist and postmarxist definitions built on Laclau (2005); Tuğal 2021.

Combining the minimal definition with more substantial categories of analysis not only thickens the analytic concept but also increases its theoretical richness. Brubaker (2017) criticised the minimal definition for only capturing his proposed vertical dimension without acknowledging the horizontal distinctions between “the people” and outside groups. For example one thickness-increasing addition would be that left-wing populism defines “the people” economically or politically against threats like globalization and imperialism, while right-wing populism defines them culturally or ethnically against outside groups and “internal outsiders” perceived as not belonging to the nation (Brubaker 2017). Engesser et al. (2017) use this thick definition to explore how they are used in a social media context and found that in Facebook and Twitter posts not one post contained all dimensions but that populist ideology was “fragmented” across posts. The more terms are added and connected the stronger and thicker the assumed ideology gets. Brubaker (2017) for example researches 5 different elites, more outsidegroups and the heartland. This comes with the caveat that the more the concept of populism is enriched the more it “lumps together” disparate political projects with disparate social bases and modes of action (Brubaker 2017; Medzihorsky and Lindberg 2024).

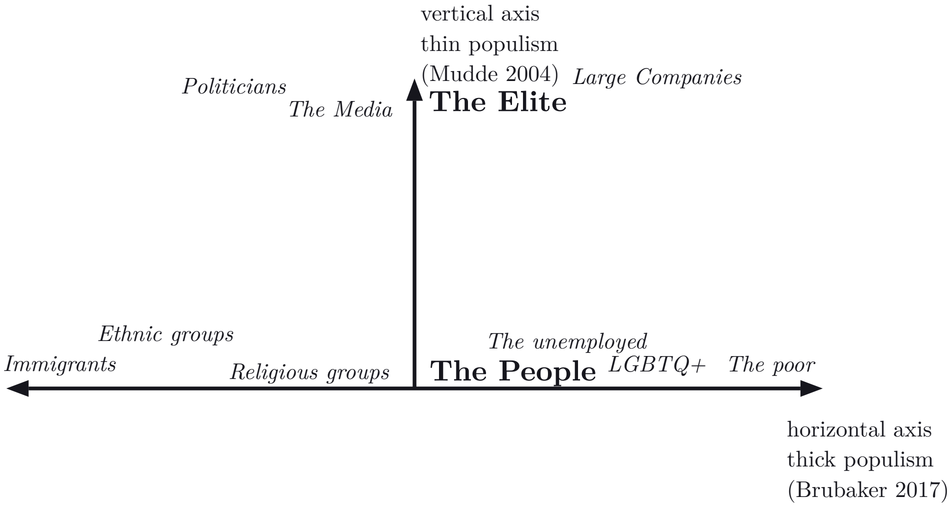

Figure 1: Vertical and horizontal dimensions of populism

After Mudde (2004) and Brubaker (2017). The vertical axis captures the thin, minimal definition: “the people” constructed against “the elite”. The horizontal axis captures Brubaker’s thick extension, in which “the people” is additionally constructed against horizontal others. The labels along the axis are illustrative examples of such horizontal others; their position on the axis is not meaningful and the horizontal dimension is not equivalent to a left–right axis.

But is populism really such a densely connected set of concepts, aka an ideology? And is it democratic or antidemocratic? While those putting left and rightwing populism on one axis do not necessarily assume that they are part of the same dimension they use it to enrich populism in order to bring it closer to an ideology. Yates and Mondon (2026) argue that on an ideological level there is no right-wing populism. Leftwing and rightwing populists disagree on the meaning of the “people” they invoke. The leftists say “plebs”, the rightwing say “ethos” when saying “the people”. While left wing populism aims at increasing marginalized groups visibility, right wing populism extends the privilege of already visible groups and leaves the oppressed and exploited oppressed and exploited (Yates and Mondon 2026). For rightwingers populism is then not part of an ideological core but merely a strategy for gaining votes. They abuse populisms ability to unify different cleavages splitting the critique of elites for different reasons (Urbinati 2019). In short, even though thin vertical populism is often enriched with the left and right dimension only leftwing populism can be ideological. The distinction becomes clearer when populists gain power. Populist leaders in power either (1) reaffirm their pro-people identity, remaining in a permanent electoral campaign or (2) change rules to strengthen their decisionmaking power (Urbinati 2019). While the populists continue to use propaganda facist populists revoke checks and balances (Urbinati 2019). Populist ideology arrives at a so called “partyless democracy” where the people represent themselves through populist protests and parties openly rule only for their own good (Brubaker 2017; Kriesi 2014; Urbinati 2019).

Luhmanns Systemtheory provides a contrasting image to the relation between populism and a democratic state of the different scholars. For Luhmann democracy is not making all decisions participatory because then one would reduce all decisions to decisions about decisions which Luhmann calls “Teledemobureaucratization” (Luhmann 1987). This favours opaque powerstructures and insiders, just like the partyless democracy claims. In its essence democracy for Luhmann is the division of the elite into government vs. opposition which becomes the systems binary code to select which communications are considered relevant or not. What is so special about this binary code in particular is that it dissolves fundamental paradoxes that all systems with organized powerdifferences inherit (Luhmann 1987). This split only works when society is already differentiated in enough horizontal functional systems that it does not need a head of state anymore (Luhmann 1987). Society then is too big and previously relevant systems have emancipated themselves from their role, sustaining themselves through autopoiesis. Descriptively, this overlaps with the diagnosis of the powerless democracy, even if Luhmann’s framework remains value-neutral where Urbinati’s is normative. “elite” here is a functional designation, those holding or contesting office, not a moral indictment. Populism’s elite/people distinction reintroduces a moral coding, pure vs. corrupt, into a political system whose code is government vs. opposition. From a Luhmannian perspective the thin populist ideology therefore does not naturally arise from the code but presses against it, as an attempt to override the functional split with a moral hierarchy. Populists achieve this by replacing the structural code government–opposition with their elite–people claim.

Abts and Rummens (2007) do not see the core issue in morality. While they see democracy as open and diverse populisms claim of homogenous people stands against that. While there are valid critiques against the parliamentary system populists target the system itself. For example on social media populists make the avoidance of “main stream media” part of their anti-elite attitude (Stier, Siegers, and Breuer 2025). Populists want to free democracy from its constitutional restraints and establish an immidiate partyless democracy with a direct leader representing the people (Abts and Rummens 2007). They ignore that constitutional constraints are needed to preserve the diversity and ability to replace the elite (Abts and Rummens 2007). Populist movements are paradoxical because they manifest themselves in an “undemocratic” system providing them with a stage while they themselves follow proto-totalitarian logics (Abts and Rummens 2007).

At this point a distinction between an antidemocratic and democratic populism can be made. The first one stays within the system and sees the people as diverse. In latin americas progressive populism labour unions stand against the decomposition of the party system from neoliberal movements (Brown 2020). The second one sees the people as homogenous, aims at replacing the democratic system, the split of government–opposition itself to establish a partyless anarchy in which the representant of the homogenous mass in power. In the end the deciding factor is the logic under which the movements function and what type of system they want to establish – the idealtypes are now laid out. While the thick ideology includes distinction against more horizontal groups or more systemrelated elites it is not inclusive or diverse and therefore by itself antidemocratic. The thin type is antidemocratic in the sense that it sees the people as homogenous, but when the people are invoked as diverse “plebs” instead of “ethos” one can also imagine a democratic populism that increases “plebs” representation within diversity-preserving constitutional constraints (Brown 2020; Yates and Mondon 2026).

2.2. Populism and Social Media

The majority of social media populism studies understand populism as a form of communication since they analyse it on a social media platform connecting all of its users.

Digital media use is consistently associated with higher populism, with causal evidence for far-right support and even ethnic hate crimes in democratic and authoritarian regimes (Lorenz-Spreen et al. 2022). The question is not whether social media matters but what it does structurally. The common explanation is affordance. Social media is cheap, viral and engagement-driven, exploited by populists for demagogic, anti-establishment and people-praising messaging including name-and-shame strategies (Gil De Zúñiga, Koc Michalska, and Römmele 2020). Cassell (2021) confirmed this through qualitative coding of populist leaders tweets across Latin America and Europe, where populist frames outperformed pluralist, technocratic and neutral tweets in engagement metrics as likes and retweets. Yet this engagement advantage is asymmetric. Dutch populists reciprocate interaction on Twitter less than non-populists (Jacobs and Spierings 2019). Populist communication mobilizes attention upward without distributing it downward. Politicians also adapt to platforms rather than to policy. Stier et al. (2018) used a Bayesian semi-supervised single-membership language model to show that candidates use Facebook and Twitter for distinct purposes, discussing campaign events and platform-specific topics rather than policy. The audience is no longer a mass but chooses its broadcaster, which makes strategic tailoring rational. Hu (2024) explores the engagement of political tweets labeled by topics and finds few follower of politicians using selfexpressive, argumentative as well as mobilizing language. These patterns aggregate at the group level. Stier et al. (2017) showed AfD and Pegida share Facebook userbases, mutually like each other and converge on topics like crime, sexual assaults, EU referenda and “state and the people”. Stier et al. (2025) found through linked survey and webtracking data that radical right populists and their supporters avoid public broadcasters and expose themselves to alternative channels. “How to talk about and select media sources seems to have developed into a core component in the construction of a radical right group identity and a shared information ‘safe space’ for political information” (Stier et al. 2025). Group identity is built around the choice of information infrastructure itself. This ties populist communication to a legitimacy crisis rather than to a left-right axis. Analyzing 32 million tweets from parliamentary accounts in 26 countries, Törnberg and Chueri (2026) found that neither left-right nor populism alone explains misinformation spread. Both left and right-wing populists consume more misinformation, but only the rightwing believe themselves better informed. The radical right has built an alternative media ecosystem in symbiotic relationship with attentioneconomic platforms (Törnberg and Chueri 2026). Consistent with the Luhmannian reading above, once a functional system emancipates itself from its societal task it begins to produce its own environment to keep itself alive.

To summarize, populism is a style of communication that constructs the divide between the people and the elite with usually rule-destabilizing consequences. Even though we can not know the individuals motivations or reasons behind usage of populist rhetoric, what is sure is the role politicians assign themselves to and what being a politician implicates. From a point of political communication politicians use this language in order to put themselves into a position to mobilize and govern.

2.3. Language and the Meso Level

Up until now it was established that populist language is used by politicians on social media platforms and that it has been connected to certain language as well as engagementmetrics.

But engagementmetrics only report on the reactions of single users, but users interact, especially on social media platforms. Those interactions of single users are conceptually not independent since social media recommender algorithms use collaborative filtering. This is especially well known for Twitter since X open sourced the architecture of the platforms recommender algorithm (X 2026). Multiple steps in the recommender algorithm are enriched with collaborative filtering techniques like SimClusters, the UTEG (User-Tweet-Entity-Graph) and the knowledgegraph (X 2026). Through this algorithm Twitter shapes the interactions on its platform. But not only from a technical perspective groupdynamics between users should be considered. One of the oldest sociological idea is that antagonistic groups shape each other. "… that through [the dispute] not only does an existing unit concentrate itself into a more energetic unit, and radically eliminate all elements that could blur the sharpness of its boundaries against the enemy – but that [the dispute] brings together persons and groups who otherwise had nothing to do with each other." (Simmel 1908:251). Tajfel et al. (1971) found ingroup favourtism in experimental settings even when common good strategies were available at a small cost unchallenged by individul benefits.

For Simmel a group needs at least three persons (Simmel 1908:Ch2). This is mainly because when one person leaves the group it doesnt automatically dissolve like if in the case of only two persons. For three persons the group can sustain itself if a person leaves the group. That a group emerges in the real world two processes are needed, homophily for the group cohesion and repulsion of others for defined group boundaries (Stadtfeld, Takács, and Vörös 2020). Pure attraction only explains a groups expansion while a heterophob repulsion mechanism creates stable groups by defining its boundaries. Stadtfeld et al. (2020) show this by calibrating a stochastic actor oriented model to 479 students from 13 schoolclasses in order to simulate how friendship and dislike networks emerge. In Twitter retweet networks language is used as a marker for opinion-based group formation (Morselli et al. 2026). Opinion based group formation requires opinions as identity markers that individuals use to transition from holding an opinion to selfcategorizing themselves through the opinion (Morselli et al. 2026). In linguistic theory the identity markers changing meaning for the ingroup is described through the distinction between esoteric and exoteric language (Wray and Grace 2007). Esoteric language is specialized in the sense that it is used for ingroup communication and outsiders can not understand it. Exoteric language is more self-explanatory and is used for communication with outsiders (Wray and Grace 2007). The coevolution of language and social groups is already activly researched in the science of science. For example Schmitz, Schmidt-Wellenburg, and Volle (2025) trace the evolution of scientific groups in the US and German sociology through a stochastic blockmodel of a multilayer network operationalized through co-word usage, shared citations and co-authoring.

The social media studies so far were mostly largescale observational studies. Experimental studies in controlled contexts can give closer insight into the social mechanics of online group/discussion formation. Oswald, Schulz, and Lorenz-Spreen (2025) paid participants to engage with each other about political topics in Reddit forums they moderated. Participants additionally filled out surveys regularly over a period of four weeks. When users perceived a discussion to be toxic they did not engage in the discussion. The users who engaged, engaged more when the discussion was polarized and toxic (Oswald et al. 2025). With a Luhmannian read the results can be generalized to the formation of a code in the discussion as a social system. Users perceive the community as positive/negative according to their own standards and start with the second stage of communication: "Mitteilung" (Luhmann 1984:Ch.4). While different persons engage according to their own evaluative codes their communication creates thematic and meaning structures (Luhmann 1984:Ch.4). The social-discussion-system starts to reproduce itself. If now the discussions would start to be selfdescriptive the system starts making a distinction between itself and its environment and emerge as a differentiated unit itself (Luhmann 1984:Ch.4). Oswald et al. (2025)s experiment also include the persons being driven away from the discussion through e.g. the evaluation of a toxic environment. With the Luhmannian read, outside of the experimental context they would put their attention somewhere else, creating new opportunities for social systems to emerge, driving again others away. On one hand the discussion becoming more homogenous reduces complexity, but it also increases comprehensible complexity Luhmann (1984). Higher connectivity (Netness) around a certain category (Catness) is equivalent to a higher amount of organization of a group (Tilly 1978:63; White 2008). Parties are aggregating members loyal to the category, aka political interest and communicative practice, aka code, making it distinct from its environment (Tilly 1978:76). Parties are then social movement because they are mobilizing others (Kusche 2016; Tilly 1978). Intergroup violence is not motivated by the out-group hate but through the spiraling "cheap talk" happening inside the in-group (Mäs and Dijkstra 2014).

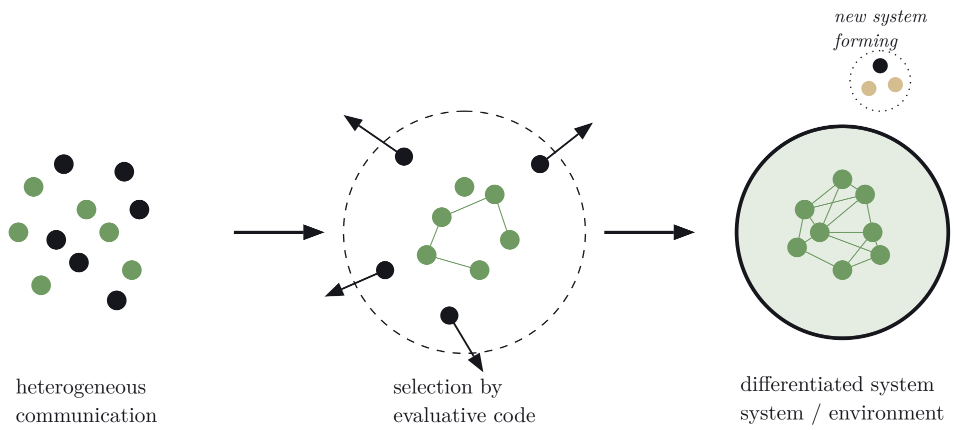

Figure 2: Three-stage account of group emergence in online discussions

After Oswald et al. (2025) read through Luhmann (1984) and Tilly (1978). At the first step communication is heterogeneous: users with divergent evaluative codes (positive in green, negative in pink) all participate in the same discussion. At the second step the system selects: users who evaluate the discussion negatively disengage, while those who evaluate it positively intensify and form initial ties. At the last step a homogenized code reproduces itself, ties densify (Tilly's netness around a shared catness), and the discussion makes a distinction between itself and its environment, emerging as a differentiated social system. The displaced users seed new systems elsewhere, restarting the cycle.

Populism (Elite/People) is the reproduction of the democratic system (Government/ Opposition) through the social movement as a subsystem of the political system. While the representative democratic system distinguishes between rulers and ruled, populism does the same, realizing its potential for mobilization. Democratic system select communication through this code (Luhmann 1987) and by reproducing it the social movement attempts to place itself in a position of being a valid democratic group. This explains why right-wing populists use this distinciton as a mobilization strategy even though it contradicts their elitist ideology (Yates and Mondon 2026). Being for the people as an ingroup is very universal since every voter already is part of the constructed ingroup from the perspective of democracy. Communication with this distinction has great potential for "Anschlusskommunikation". It drags everyone in the political arena, puts citizens at the negative-opposition side of the Government/Opposition distinction inviting them to participate in the movement empowering them as political citizens.

A more active engagement community should therefore manifest itself through more observed ties around the populist politician.

Or more precisely for the given Dataset:

German MPs who use populist rhetoric have higher interconnected alters in their reply ego networks compared to MPs using less populist rhetoric.

Engagement networks are special in the sense that they do not reflect who agrees with whom but who interacts with whom. Morselli et al. (2026) found that close individuals in the twitter retweet network can have different opinions. Even though on a linguistic basis at the first glance it seems unreasonable, but we can imagine the extreme case. A user is always coming back to a politicians tweets and its fanbase to share their disagreement. Are they part of the group or not? The person is not part of the fanbase but definitly part of an engagement community where individuals continuously show up in the same place and position, no matter why.

3. Data and Methods

This section describes the data, measures, and analytical strategy used to examine how populist rhetoric by German MPs shapes their engagement communities on Twitter. After introducing the dataset and collection procedure the three step analysis is described.

First, as the main part of a text analysis populism is operationalized as a three-dimensional construct of People Attitude, Elitist Attitude, and Antagonism, scored by a large language model through a structured annotation prompt. Second, the construction of reply threads, the user-level reply network and the exploration of their structure are detailed. Finally in a synthesis, politician ego networks are combined with the populism scores produced by the text analysis. Linear Regressions with their controls, as well as a robustness check and further exploration of the ego networks is described.

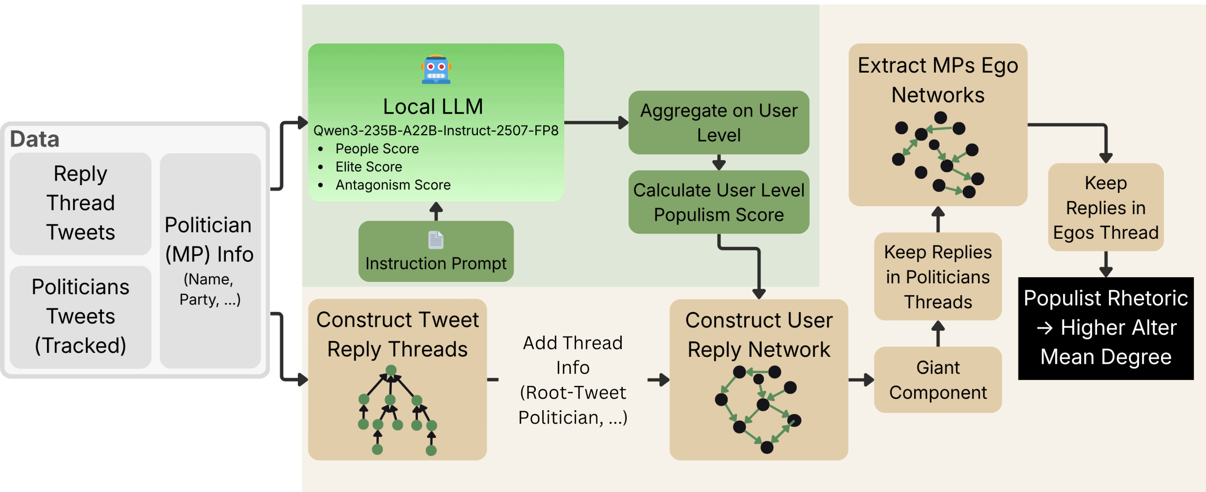

Figure 3: Diagram describing conducted Analysis

3.1. Data and Operationalization

The dataset used in this study was kindly provided by Dr. Armin Pournaki from the Max Planck Institute for Mathematics in the Sciences, Leipzig. The dataset holds tweets and replies directed at German members of parliament as well as their retweets and referenced tweets, collected over a one week from February 7 to February 14, 2022. Only the MPs tweets and their replies are analysed since the reply network best captures social interactions of engagement communities. Data were obtained via Twitter's streaming API using two parallel strategies. The follow stream tracked retweets of and replies to MP-generated tweets, while the track stream captured tweets mentioning MP handles (e.g., "@username"), including direct and nested replies. Protected accounts were excluded. Both datasets were combined and deduplicated. From the raw data, reply-threads were constructed by linking tweets through directed reply chains. These interactions were then aggregated to a user-level reply network, weighted by the number of exchanges. Thus, the dataset enables an analysis of interaction patterns between politicians and their engagement communities on social media. To deliniate the set of politicians and add their name and party information, a list of parliamentary members of the 19th, 20th, and 21st German Bundestag is obtained from the Bundestags Webarchiv and added to the original dataset through their twitter-account links (Bundestag 2026). This ensures that the analysis accounts for politicians active in the legislative periods preceding, during, and following the timeframe of the collected tweets. The complete dataset consists of 693 015 Tweets in the twitwi format (medialab 2026).

German political news from 7–14 February 2022 were dominated by the Omikron wave's peak and the escalating Russia-Ukraine crisis. As Russia massed troops for what would become Europe's largest military offensive since WWII, Chancellor Scholz met President Biden to discuss Germany's Nord Stream 2 dependency. Domestically, Scholz's coalition split over a general vaccine mandate, with FDP Justice Minister Buschmann deeming it constitutionally dubious and proposing mandatory physician consultations for unvaccinated adults as a softer alternative.

Current studies almost uniformly base their understanding of populism on Mudde (2004)'s definition of populism as "two homogeneous and antagonistic groups, 'the pure people' versus 'the corrupt elite', and which argues that politics should be an expression of the volonté générale (general will) of the people." (Mudde 2004:542). While all operationalizations include a "pro-people" and "anti-elite" one of multiple third dimensions is often implemented as well like the inclusion of anti-pluralist attitudes, the distinction between leftwing and rightwing populism or agitating against horizontal outgroups like minorities (Aalberg et al. 2017; Castanho Silva et al. 2020; Meyer et al. 2025). Populist Attitudes are not only measured through surveys (Castanho Silva et al. 2020), but also through observational studies of political discourse on social media (Meyer et al. 2025) leveraging LLMs.

This study operationalizes the core dimensions of Mudde (2004)'s definition, People Attitude, Elitist Attitude, and Antagonism, by instructing a large language model through an annotation prompt (see Appendix). Each dimension is defined with explicit scoring anchors: People Attitude and Elitist Attitude are measured on bidirectional scales from –3 to +3, where positive values indicate support for and negative values indicate opposition to the respective group, while Antagonism is measured on a unidirectional scale from 0 (no divide) to 6 (existential threat), with labeled thresholds distinguishing dissatisfaction (1–2), active blame (3–4), and existential threat framing (5–6). The prompt leverages chain-of-thought style reasoning and few-shot examples to guide the models annotation behaviour. To guide the models reasoning letting it question itself throughout the process lead improved the results immensely compared to hard rule-based checks.

The combination of dimensions is necessary to capture all parts of Mudde (2004)'s minimal definition visualized in Figure 1. All parts are necessary because anti-elitism is its own concept, it might be a good proxy for populism but does not capture the essence of the constructed ingroup. Only observing pro-people attitudes does not account for the delineating differentiation of the people from the elite. Both groups are rhetorically presented as actors that and are emotionally evaluated (Klingelhöfer 2026). Both an ingroup and outgroup are needed to induce a lasting social structure (Stadtfeld et al. 2020). The strength of the motivated moralization and emotionalisation of the groups as actors is represented as a rhetorical distance constructed between the groups through the `antagonism score`. Only the combination of all three makes up the minimal populism found in the literature. By that the system prompt directly translates the definition into an operationalized measure. Populism for the analysis is aggregated at a user level, not only because politicians as users are the unit of analysis, but also because populism on social media is fragmented (Engesser et al. 2017). Politicians share crumbs of their ideology at different places and only the combination can indicate a communicative rhetorical strategy.

The prompt follows the structure described by Liu et al. (2026). It begins with a role definition and a pre-analysis check if the text carries any content other than just a link or a user mention. It proceeds by defining "People Attitude", "Elite Attitude" and "Antagonism". The definition of "the people" is restricted to a broad ordinary majority and explicitly excludes named individuals, lists of specific persons, and narrow subgroups unless the text frames them as standing in for the general public. Similarly, elite criticism is only scored when the target is a generalized powerful class rather than a single individual or a specific policy disagreement. In the last major section the prompt invokes the chain-of-thought before assigning scores, the model must produce a holistic redescription of the post's rhetorical strategy, an actor-by-actor analysis that classifies each referenced person or group by scale (individual, institution, or generalized class) and dimension-specific explanations that articulate the reasoning behind each score. At each step, the model is asked to consider alternative readings and flag its confidence as "low" when a reasonable coder could disagree. Three few-shot examples are included to calibrate the model's decision boundaries: a strongly populist post with high people, elite, and antagonism scores, a non-political post that should receive all zeros, and an ambiguous post where institutional criticism could plausibly be read as either targeted policy dissatisfaction or broader anti-elite attitude. This example structure is designed to discourage binary classification tendencies and encourage the model to use the full range of each scale.

The prompt is included as a systemprompt and appended with the to be annotated tweet. The model outputs its reasoning and scores in a json format.

After coding the scores are combined. The three dimension scores are denoted $P \in [-3, +3]$ (People Attitude), $E \in [-3, +3]$ (Elitist Attitude), and $A \in [0, 6]$ (Antagonism). The tweet-level populism score is

$$\text{Populism} = \begin{cases} (P - E) \times A & \text{if } A \geq 1 \\ P - E & \text{if } A = 0 \end{cases}$$

The subtraction captures the joint rhetorical direction that is positive when the people are elevated and elites denigrated yielding a theoretical range of $[-36, +36]$. Politicians using populist strategies on social media do not have to adress all populism dimensions in one tweet but can be anti-elitist in one and pro-people in the next. Therefore the combination on the user level is achieved by averaging each dimension separately before recombining. Let $\bar{p}_u$, $\bar{e}_u$, and $\bar{a}_u$ denote the per-user means. The user-level score is

$$\text{Populism}_u = \begin{cases} (\bar{p}_u - \bar{e}_u) \times \bar{a}_u & \text{if } \bar{p}_u > 0, \bar{e}_u < 0, \bar{a}_u > 0 \\ \bar{p}_u - \bar{e}_u & \text{if } \bar{p}_u > 0, \bar{e}_u < 0, \bar{a}_u = 0 \\ 0 & \text{otherwise} \end{cases}$$

The gate zeros out users whose average rhetoric is not simultaneously people-affirming and elite-critical, operationalizing Mudde (2004)'s tripart definition at the actor level. Aggregating dimensions before recombining allows a user to distribute their anti-elite and pro-people attitudes across different tweets.

3.2. Textanalysis

Computationally methods driven text analysis is based on linguistic concepts. A corpus of documents is divided in tokens as semantically meaningful units of analysis which are often words. The simplest text classification techniques are usually wordfrequency based dictionary methods. Some of which already mechanistically take context of wordappearence into account (Ribeiro et al. 2016). While simple machine learning classifiers like naive bayes models are statistically more elaborate, deeplearning models are able to integrate interactions between tokens allowing for even better predictions. But deep learning architectures like convolutional neural networks trade variance for increased local bias through usage of filters (Sohil, Sohali, and Shabbir 2022). Transformer architectures implement modeling of long-range dependencies by implementing self attention on a micro level weighing the importance of every individual input token for other each individual input token (Cordonnier, Loukas, and Jaggi 2020). Qwen3-235B-A22B-Instruct-2507-FP8 is using more elaborate attention mechanisms like grouped query attention (Yang et al. 2025). The model also implements expert segmentation, is multilingual and has 8-bit floating weights (Yang et al. 2025). To run the model most efficiently on the HPC Capella Cluster of the TU-Dresden the vLLM framework was used through python to make use of virtual memory PagedAttention techniques (Kwon et al. 2023).

Output quality of large language models is highly dependent on prompt design (Liu et al. 2026). Prompt engineering is conducted iteratively by drawing small random samples of tweets from the corpus and classifying it using an initial prompt. The resulting reasoning steps contained in the LLMs JSON output, like the actor analysis for example, are then qualitatively assessed with regard to coherence and later label accuracy. Based on identified shortcomings, the prompt is systematically adjusted and the process is repeated. Early versions of the prompt were structured in the style described by Liu et al. (2026). Later additional rule based checks were integrated and the coherence of the phrasing was adjusted. The rule based reasoning guidence was replaced for the final prompt with reflective questions prompting the LLM to question its judgements and certainty. This was related to a large increase in reasoning and output performance. These adjustments were done with a tiny subsample of the dataset. External validity is tested after arriving at a sufficiently well performing prompt. The expert annotated training dataset of Bundestag speech sentences, used to fine tune the PopBERT populism classification transformer, serves as a final validation of the prompt (Erhard et al. 2025). The dataset contains 8795 sentences of Bundestag speeches with binary annotations of five experts on the people-centrism and anti-elite dimension (Erhard et al. 2025). After annotating the validation dataset locally with the developed systemprompt and Qwen3-235B-A22B-Instruct-2507-FP8 accuracy

$$\text{Accuracy} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\left[\text{rater}_A^{(i)} = \text{rater}_B^{(i)}\right],$$

as well as F scores

$$\text{Precision} = \frac{\text{both rated positive}}{\text{both rated positive} + \text{only predicted positive}}$$

$$\text{Recall} = \frac{\text{both rated positive}}{\text{both rated positive} + \text{only reference positive}}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

for pairs of expert raters with each other and the LLM annotations are calculated. For validation the bidirectional rating scale of the LLM is collapsed to binary labels matching the experts labels of pro-people and anti-elite. Pairwise calculations are possible because the F1 score is not symmetric for swapping labels but it is symmetric for swapping raters (see Appendix).

3.3. Network Analysis

The reply threads are constructed by treating tweets as nodes that are linked through directed replies. A component in the constructed thread network is then a replythread with one rootnode that is only receiving links marking the direction of a reply. Some reply chains trace back to MP tweets posted before the collection period, so the root node has no row in the dataset. Matching the target tweets user information against the politician table recovers their authorship, recovering 2,631 of 89,561 threads (2.9%). Threads root tweet information like politician with e.g. party and followerinformation, are then stored within the reply tweets of the respective thread and as well added to each tweet in the reply network.

The main research question requires a network of interactions plausibly representing an engagement community. A weighted user centric reply network with users as nodes and replies as directed links is constructed. The original tweet-based dataset holding politician and threadinformation served as the basis for construction with its `user_id` and `to_userid` variables. The resulting network is directed with edgeweights based on replycount. It holds 81 295 user-nodes connected through 239 502 reply-links distributed among 1865 components. A giant component holds (77 194) 94% of all nodes. 1207 users are contained in components of size 2, one original user, one replying user, as the threadsize holding the second most nodes. The second largest component held 41 (0.05%) users (see the table in appendix). The network analysis is based on the giant component to make computation more feasible leading to a loss of the network periphery without any actual structurally relevant threads.

Tweet-level populism dimensions (people scores, elitism scores, antagonism scores) are aggregated per user by computing weighted means across each user's tweets, then combined into a userlevel composite populism score. These user-level scores are added as vertex attributes to the reply network.

Because the analysis focuses on politicians reply threads of the 19th, 20th and 21st German Bundestag all replies not started under those politicians root-tweets are removed and the network is cleaned of isolates. This reduces the networks size by half, but makes sure only politicians acting in their political role are captured.

A descriptive exploratory analysis describes first the results of the textanalysis and second the networkstructure. Following that, first the most relevant descriptive network statistics are calculated. Additionally the reply network structure is explored through a visualization using the Distributed Recursive Graph Layout (Martin, Brown, and Wylie 2007). The Distributed Recursive Graph Layout (DrL) lays out a graph by applying repulsion and attraction forces between nodes to prevent overlap and keeping connected nodes close. Beginning with randomness so nodes can move freely and avoid poor configurations, gradually reducing movement until positions stabilize. Once positioned, spatially close nodes are merged into representative summary nodes producing a coarser version of the graph. The cycle of coarsening and repositioning repeats until the graph is sufficiently small to lay out directly. Finally the process is reversed, expanding each simplified graph back one level at a time, using the prior layout as an initial arrangement.

3.4. Ego Network Analysis

After the extensive exploration and descriptive analysis of the dataset, the structure of politicians ego networks is compared to answer the main question about local engagement communities. Mean alter degree is used as the primary measure of alter interconnectedness, capturing the average number of connections each alter maintains to other alters within the ego network. While density would be a more established measure in usual cases it is inversely related to network size, making comparisons across ego networks of different scales unreliable. As a robustness check, the fragmentation ratio is additionally tested since it relates the number of connected components to the overall network, providing a complementary perspective on whether alters form a cohesive neighborhood or fragment into isolated clusters.

$$\text{Fragmentation} = \frac{\text{component count}}{n_{\text{alters}}}$$

For controlvariables the politicians (ego) degree and followercount control for the popularity of a politician while mean thread size of the politicians seeded threads controls for the depth of conversational engagement elicited by that politician. Ego degree captures structural prominence in the reply network, followercount proxies platform-level visibility, and mean thread size accounts for the possibility that longer threads mechanically increase the chance of alter-alter interaction simply by providing more opportunities for replying.

Two OLS regression models are estimated. The baseline model regresses mean alter degree on a binary populism indicator derived from the composite populism score. The full model adds ego degree, follower count, and mean thread size as controls. Comparing the populism coefficient across both models reveals how much of the observed relationship between populist rhetoric and alter interconnectedness is attributable to confounding structural and visibility differences between politicians.

As a robustness check, the fragmentation ratio is tested as an alternative dependent variable using the same specification as the full model. Where mean alter degree captures the intensity of alter interconnectedness, fragmentation captures its inverse: the degree to which the ego network consists of isolated clusters. A consistent result across both measures strengthens the claim that populist rhetoric is associated with more cohesive engagement communities instead of an artifact of the chosen operationalization.

The outlined regression models only control for ego attributes, but the role of alters is left open. To explore the interaction between alters further, the egonetworks are summarized by their edges. The resulting table holds all edges of all ego networks. To meaningfully compare edges across ego networks of varying size and alter composition, each observed edge statistic is contrasted against a baseline, computed of random-pairings within the same ego network. Following ERGM conventions, two statistics are calculated for each continuous alter attribute (follower count, tweet count, populism score): the mean absolute difference between connected alters (`absdiff`, $\overline{|x_i - x_j|}_{\text{obs}}$) captures homophily, while the mean sum across edges (`nodecov`, $\overline{x_i + x_j}_{\text{obs}}$) captures preferential activity. The same quantities are then evaluated over all possible unordered alter pairs in the ego network ($\overline{|x_i - x_j|}_{\text{rand}}$ and $\overline{x_i + x_j}_{\text{rand}}$), yielding the expectation under uniform random pairing. The observed value is finally expressed as a relative deviation from this baseline, $(\text{obs} - \text{rand}) / \text{rand}$. For `absdiff`, a negative deviation signals that connected alters are more similar than a random pairing of the same alters would produce. For `nodecov`, a positive deviation indicates that edges disproportionately link high-attribute alters. By holding the alter pool fixed within each ego, this normalization isolates the structural signal carried by the observed edges from variation in alter composition and ego network size, making the resulting deviations comparable across politicians. Both statistics are summarized by median and mean across populist and non-populist ego networks to identify possible systematic group-level patterns in tie formation.

4. Results

4.1. Textanalysis

To rate the tweets populism scores a system prompt was developed as described in Section 3.2 instructing the LLM with the operationalization scheme of populist rhetoric. The prompt is validated by annotating an already expert rated dataset of sentences from Bundestagspeeches used to train the PopBERT classifier (Erhard et al. 2025). The 8795 sentences were rated by 5 expert raters on a binary scale if they contain anti-elitist or people-centrist attitudes. since the LLMs system prompt rates the same dimensions in higher resolution the output was collapsed to a binary rating for comparison. The LLM rates if a tweet contains an attitude towards the elite or the people, each on a scale of -3 to 3 where negative numbers represent negative attitude towards the group and positive numbers positive attitudes. An elite_score from -3 to -1 represents anti-elitism and was collapsed to 1, while all other ratings were collapsed to 0. The same was done to the people_score.

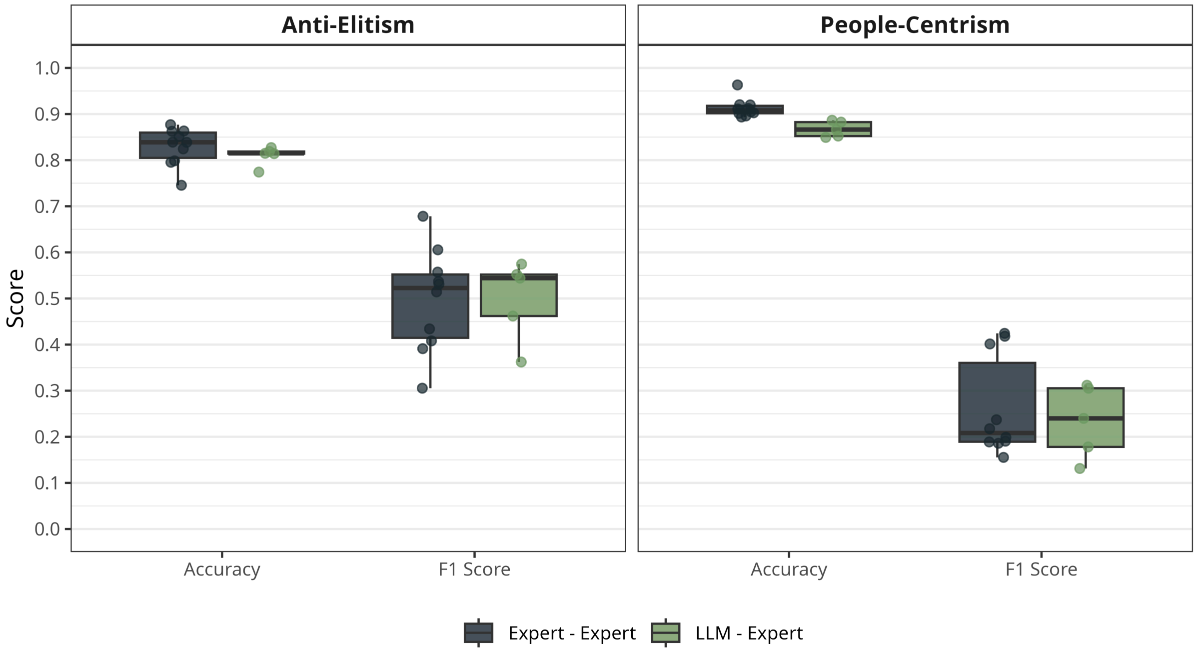

Figure 4: Prompt Performance - Pairwise accuracy and F1 scores for all expert-expert and LLM-expert rater combinations across anti-elitism and people-centrism.

Pairwise accuracy and F1 scores between binarized Qwen3-235B-A22B-Instruct-2507-FP8 ratings using the developed populism system prompt and five expert annotators from the PopBERT training corpus (Erhard et al. 2025). Each point represents one rater pair (expert-expert or LLM-expert). LLM ratings were collapsed from a 7-level Likert scale to binary scores (anti-elitism: score < 0; people-centrism: score > 0).

Figure 4 shows the pairwise accuracy measures and F1 scores between each combination of expert raters as well as each combination between the LLM and one of the expert raters. The rater-rater and rater-LLM accuracyscores are quite high for anti-elitism with a median between 0.8 and 0.9 and people-centrism with a median between 0.85 and 0.9. Accuracy is the proportion of tweets on which two raters assigned the same binary label. Since most sentences seemed to contain no anti-elitism or people-centrim the dataset was skewed which inflates the accuracy scores. The raters and the LLM have high agreement on the large count of 0 values.

To accurately assess the performance of the prompt pairwise F1-scores are calculated for each rater-rater and rater-LLM combination. It adjusts the accuracy by ignoring zero agreements and only rewards raters for agreement on rare positives. The rater-rater and rater-LLM combinations for the anti-elitism score produce a median F1 Score of slightly over 0.5, while the median F1 score for people-centrism lie around 0.2 - 0.25.

Populism seems to be a fuzzy concept where even expert raters have a hard time deciding on the strength and presence on the anti-elitism dimension but agree even less on the people-centrism dimension. What still stands out is that the LLM-rater combinations lie in the same regime as the rater-rater combinations successfully making an LLM equiped with the systemprompt as precise as an expert rater in detecting dimensions of populism in bundestagspeeches.

The Qwen3-235B LLM rated all tweets by their attitude about the elite, the people and the opposition between the two groups.

Tweet	**Populism Score**	**Party**
Das kleben einem die politischen Freunde von `@PoliticianX`, sog. #Antifas an die Fenster, wenn sie nicht gerade Artikel von `@PoliticianX` in linksextremistischen Postillen lesen. Die Aussage wird vom totalitären #innenministerium vermutlich geteilt. Das Volk soll die Fresse halten.	25	AfD
Prügelnde Bullen, Tränengas, Gewalt. Warum nochmal? Um die Verbreitung eines Krankheitserregers einzudämmen? Glaubt das ernsthaft noch jemand? Es ist längst klar, dass es hier nicht um Pandemiebekämpfung geht. Sondern um die 1% gegen den Rest. Steht auf und wehrt euch.	25	AfD
Und dieser transformierte demokratische Widerstand der Bürger wird am Ende wieder die Freiheit, die Demokratie und den Rechtsstaat herbeizwingen. “Lasst uns aus dem autoritären Linksstaat wieder einen freiheitlichen demokratischen Rechtsstaat machen!” #AfD	20	AfD
Typisch #Hannover, typisch unter #SPD-#Pistorius! #Polizei kuschelt mit #Clan-Kriminellen und #linksextremen Gegendemonstranten. Aber rein schikanösen #FFP2-#Maskenzwang (AN FRISCHER LUFT!) setzen Polizisten gegen friedliche Bürger brutal um, im Stil von “Waas guggst Du?”-Typen!	20	AfD
Gut erkannt! Aber das ist doch das Ziel, das immer währende Drangsalieren der Menschen. Durch #impfzwang #Impfpflicht Sog. #corona-Maßnahmen usw. Darum: #Spazierengehen! Dem gewissenlosen Establishment zeigen: #Impfpflichtneindanke	20	AfD

Table 1: Top-scoring tweets on the composite populism scale.

The individual dimensions are aggregated by user. By that each user is positioned in a three dimensional space of those values. Figure 5 visualizes this space. Most users lie at the zero value on all axis (better seen in Figure 17). Most users lie in bottom half of the plot, tweeting against the elite. Of those users some are tweeting also against the people, but most tweet against the elite and for the people. Antagonism is getting stronger in the direction of the bottom corners of the space. The diagonal from the origin to the bottom right region is essentialy the axis of populist operationalization. The distribution is skewed in this direction, visualizing the real populism signal. The Appendix includes a version of Figure 5 with the highlighted position of politicians.

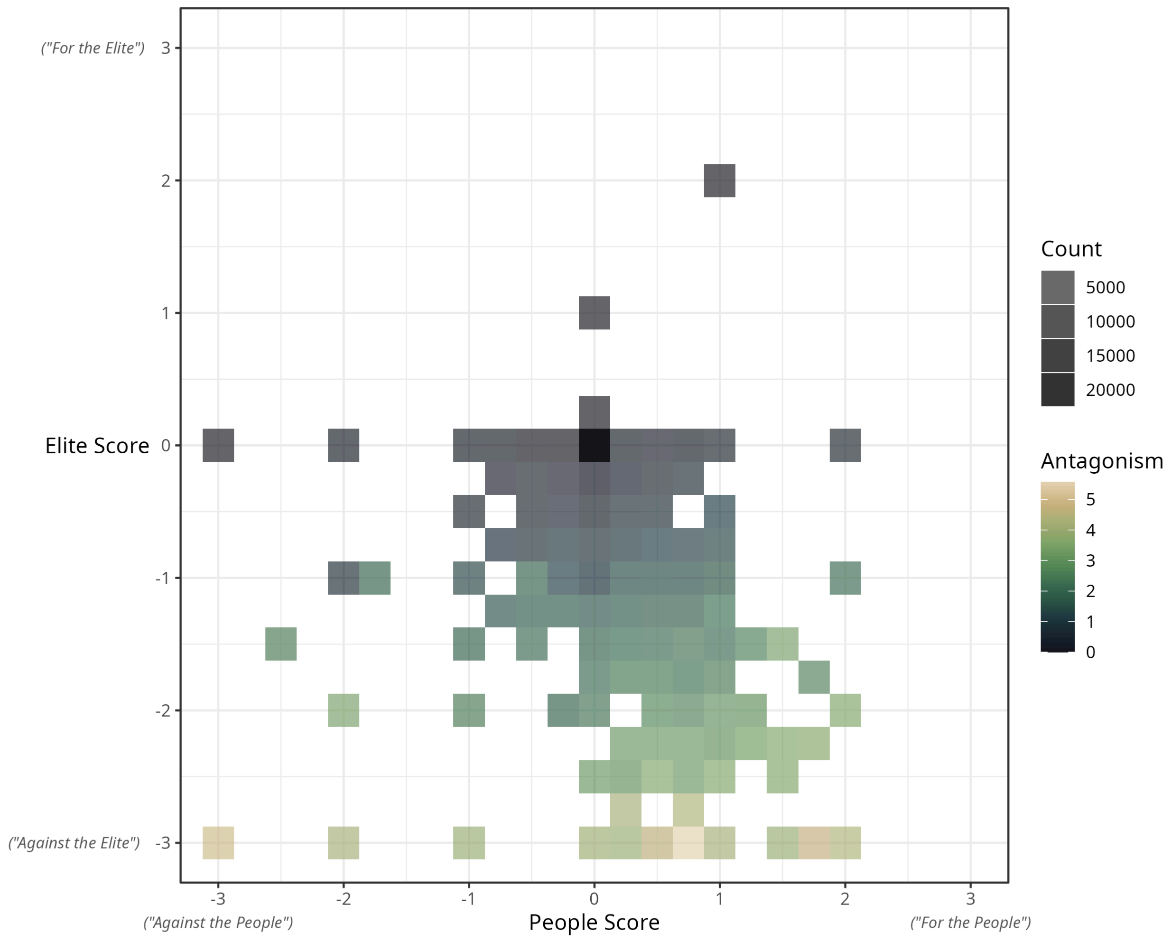

Figure 5: Distribution of populism dimensions per user in the giant component;

People score ($\bar{p}_u$) and elite score ($\bar{e}_u$) are binned into 0.25-wide intervals. Color visualizes mean antagonism ($\bar{a}_u$) per bin; opacity describes user count ($n = 29\,672$).

Figure 6 shows the relative count of tweets rated on each scale, grouped by party (sorted by mean populism) and by politician within party (sorted by populism score relative to tweet count). The plot visualizes not only politicians of the giant component as before but all politicians in the original dataset including politicians outside of the giant component or politicians only present in the retweet network. This allows interpretation of within- and between-party variation in the populism score and its components. The AfD has the most politicians using populist rhetoric, followed by BSW and Die Linke, while CDU and CSU sit in the middle and the least populist parties are the Greens, SPD, and FDP which made up the ruling coalition at the time of data collection.

Populism scores are driven almost entirely by anti-elite and antagonism components, as pro-people rhetoric is nearly absent across all parties and therefore contributes little to the composite score. Most politicians do not use populist rhetoric on Twitter, except in the AfD, BSW, and Die Linke, where a substantial share use populism in at least 1/5 of their tweets.

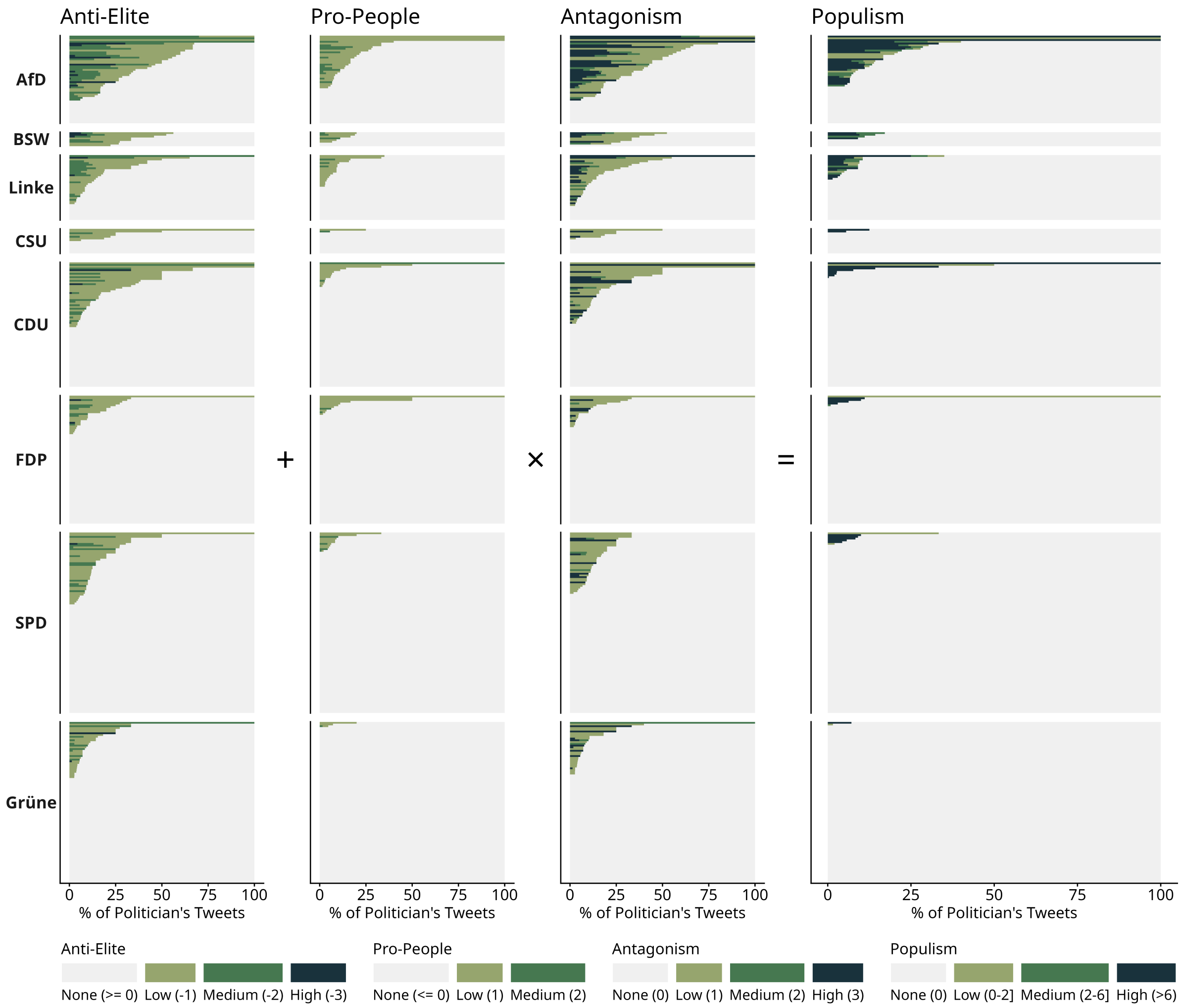

Figure 6: Populism score and its components split by parties and sorted by mean party populism scores.

Each bar is one politician in the giant component; the x-axis shows the share of their tweets at each score level assigned by Qwen3-235B-A22B-Instruct-2507-FP8 with the expert level annotation prompt. The composite populism score equals (Pro People Score + Anti Elitism Score) × Antagonism Score when Antagonism Score > 0, otherwise Pro People Score + Anti Elite Score. Dimensions visualized for direction increasing the populism score. Parties sorted by descending mean populism score; politicians sorted within party by non-zero share.

To get a deeper insight into the content of the tweets Figure 7 displays the term-frequency inverse-document-frequency correlation network colored by populism score. Tf-idf weights a term proportionally to its in-document frequency and inversely to its corpus-wide document frequency, treating concentration as a measure for distinctivness. The top tfidf values across a corpus represent characteristic key words of the entire collection of documents, while the visualized top correlated tf-idf show distinctive words that appear together potentially revealing clusters of relevant topics. The network layout is a centrality-constraint radial layout as developed by Brandes and Pich (2011). It was implemented through `layout_tbl_graph_centrality(cent = closeness(GRAPH))` (Pedersen and RStudio 2025).

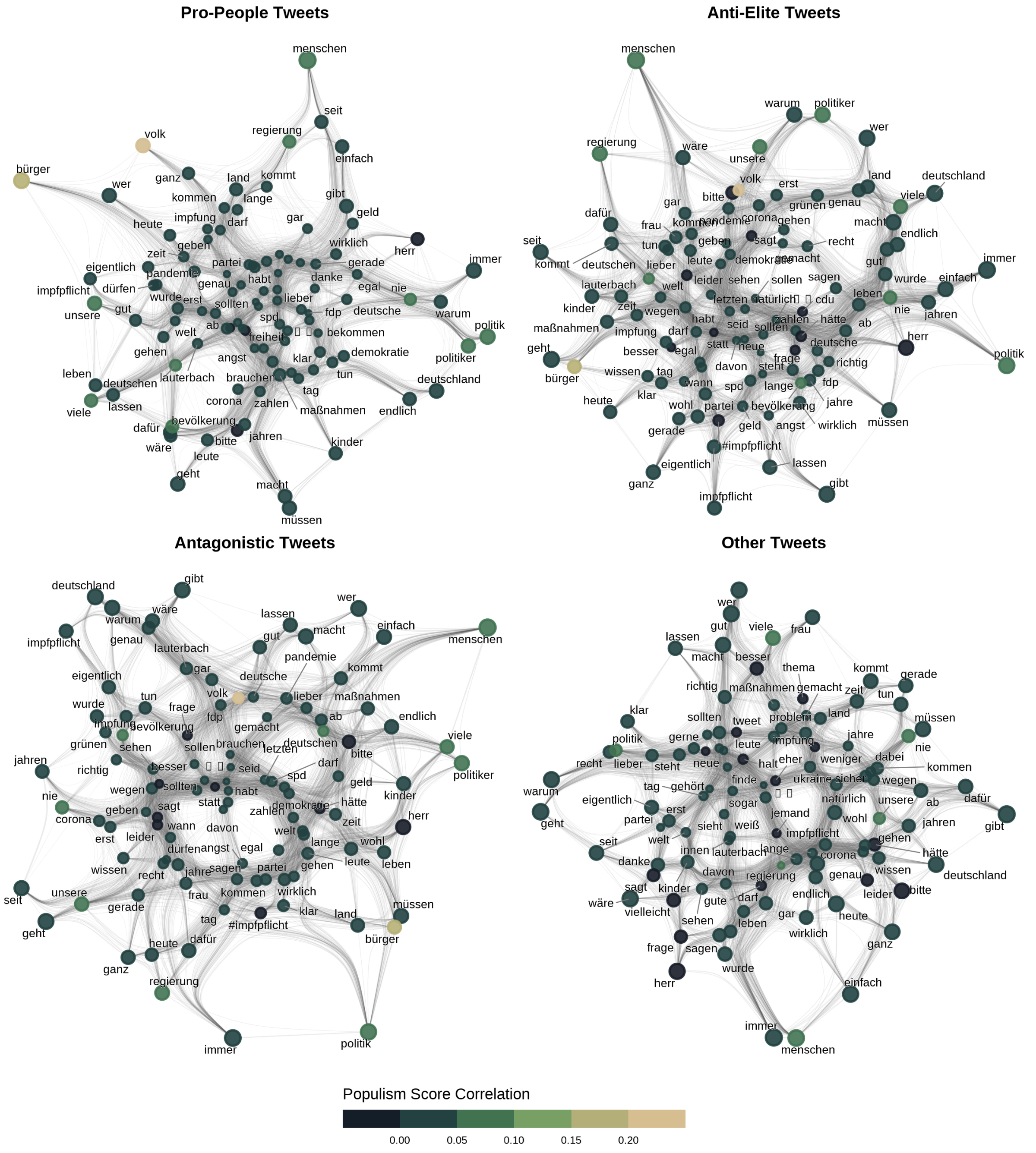

Figure 7: Top 100 tf-Idf correlated words of politicians

MPs populism score > 0; MPs populism score < 0. Nodes colored by the correlation of the original tweet with respective populism score; Network laid out with closeness-constrained radial layout (Brandes and Pich 2011). Nodes with similar shortest average distance are placed on the same circle and face the same angle by sharing common neighbors. 4024 tweets from non-populist MPs, 2275 tweets from populist MPs.

The three populist categories share an actor and in-group rim of `volk`, `bürger`, `menschen`, `politiker`, `politik`, and `bevölkerung`, but the topical vocabulary they carry differs. Pro-People tweets foreground a normative-rights register (`freiheit`, `demokratie`, `staat`, `dürfen`, `bekommen`, `geld`, `millionen`) alongside pandemic terms (`corona`, `impfung`, `impfpflicht`, `maßnahmen`, `pandemie`, `lauterbach`). The arising topics are what citizens are owed and denied. Anti-Elite tweets shift to named opponents like parties (`cdu`, `fdp`, `grünen`, `spd`), `regierung`, and `lauterbach` cluster around `politiker` and `herr`, with the same pandemic vocabulary (`impfpflicht`, `maßnahmen`, `impfung`, `pandemie`) reframed as elite failure. Antagonistic tweets combine both parties, `regierung`, `lauterbach` and pandemic policy on one side while `bevölkerung`, `bürger`, `freiheit` and affective terms like `angst` on the other. This stages the people-elite confrontation directly. The non-populist (other) tweets show none of these patterns. The topics arising are about occuring political events and political procedures. (`ukraine`, `corona`, `impfung`, `impfpflicht`, `lauterbach`) appear but without any actor, surrounded by everyday vocabulary (`finde`, `gerne`, `vielleicht`, `sogar`, `eher`, `weniger`, `tweet`). Across the populist panels `volk` stands out with its high above 0.20 populist-score correlation. It is followed by the medium correlating `bürger`, `bevölkerung` and `politiker`.

4.2. Network Analysis

The analysis now turns to the structure of the reply network as the relational substrate on which politicians and citizens meet. Table 2 summarizes this network as a directed weighted network anchored in politicians' threads as described in Section 3.3. After this filter and isolate removal, 29,672 users remain, 99.3% of whom (29,474) sit in a single connected component. The 96 residual micro-components, all of size 2 or 3, are retained for the descriptives below. Their inclusion does not affect any reported statistic at the precision shown.

Category	**Metric**	**Value**
Size	Nodes (n)	29,672
	Directed edges (m)	60,117
Degree	Density (ρ)	< 0.001
	Mean degree ($\bar{k}$)	4.05
	SD of degree (σ_k)	55.90
Directionality	Reciprocity	0.003
	Degree assortativity (r_k)	-0.066
Distance	Average shortest path ($\bar{\ell}$)	4.21
	Diameter	139
Cohesion	Clustering coefficient (C)	0.036
	Max k-core	13
	Modularity (Walktrap)	0.498

Table 2: Descriptive Statistics of the reply network.

The reply network connects users with each other through weighted edges capturing the count of replies sent to each other. It is created from reply trees connecting tweets with each other when they reply to a original tweet or another reply of that original tweet.

The giant component contains 29,672 users and 60,117 directed reply edges at a density of 0.0001, the expected baseline for largescale online networks where any given pair is overwhelmingly unlikely to be connected. Mean degree of 4.05 places the typical user at a handful of replies while the standard deviation of 55.90 is an order of magnitude larger, marking hubs that absorb and produce orders of magnitude more, structurally the politicians themselves. Reciprocity at 0.0034 shows replies flow oneway almost without exception. Citizens reply at politicians, politicians essentially never reply back. A clustering coefficient of 0.036 leaves local cohesion vanishingly thin, two users replying to the same politician hardly ever form a tie themselves and triangles are essentially absent. Degree assortativity at −0.066 is mildly negative, highdegree nodes connect to lowdegree nodes rather than to each other. Average shortest path of 4.21 against a diameter of 139 shows the hubs compress typical distance such that any two users sit only a short reply chain apart on average, even as a few peripheral paths stretch substantially longer, while the max k-core of 13 marks a small dense core embedded in an otherwise sparse periphery. Walktrap modularity of 0.498 confirms the network partitions cleanly into well-separated communities, consistent with replies clustering around individual politicians and their respective audiences.

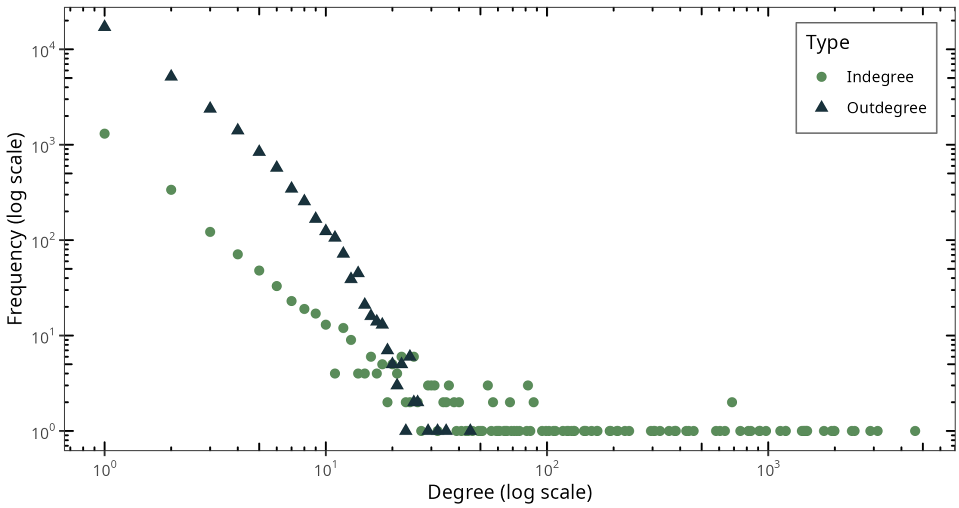

Figure 8: In and outdegree distributions of the giant component on log-log scale

Frequency of users with a given in or outdegree in the giant component (n = 68 240). Indegree counts replies received per user, outdegree counts replies sent. Both axes log scaled.

Figure 8 splits the degree distribution into its directed components and reveals two qualitatively different shapes that cross. Outdegree starts high at degree one with roughly $2 \cdot 10^4$ users sending a single reply, decays steeply across the loglog plane, and truncates near 30 with no user sending substantially more. Indegree starts an order of magnitude lower at degree one, decays more gradually, and the two curves cross around degree 10 to 20 where indegree overtakes outdegree and continues into a heavy tail reaching past 10^3. In short the majority of users has a high outdegree and low indegree, so post more than they recieve, but only few users have a high indegree. This consolidates the picture from the Table 2.

To confirm the core-periphery network structure Figure 9 a k-core decomposition of the giant component peels the least connected nodes of the network network layer by layer, revealing how nodes distribute across the depth from periphery to dense core (Seidman 1983).

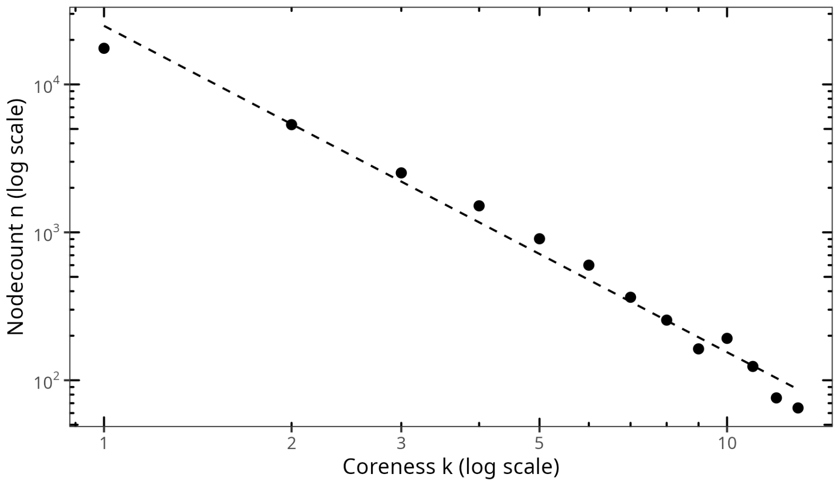

Figure 9: K-core decomposition of the giant component.

Nodecount n per coreness shell k for the giant component, on log–log axes. Each point gives the number of nodes whose maximum coreness equals k. The dashed line is a linear fit for reference. The approximately linear trend across more than three orders of magnitude indicates that shell sizes decay roughly as a power of k.

Shell sizes decline steeply and monotonically with coreness, spanning more than two orders of magnitude. The decomposition reveals a continuous gradient from a large periphery to a small dense core, without a sharp core/periphery boundary.

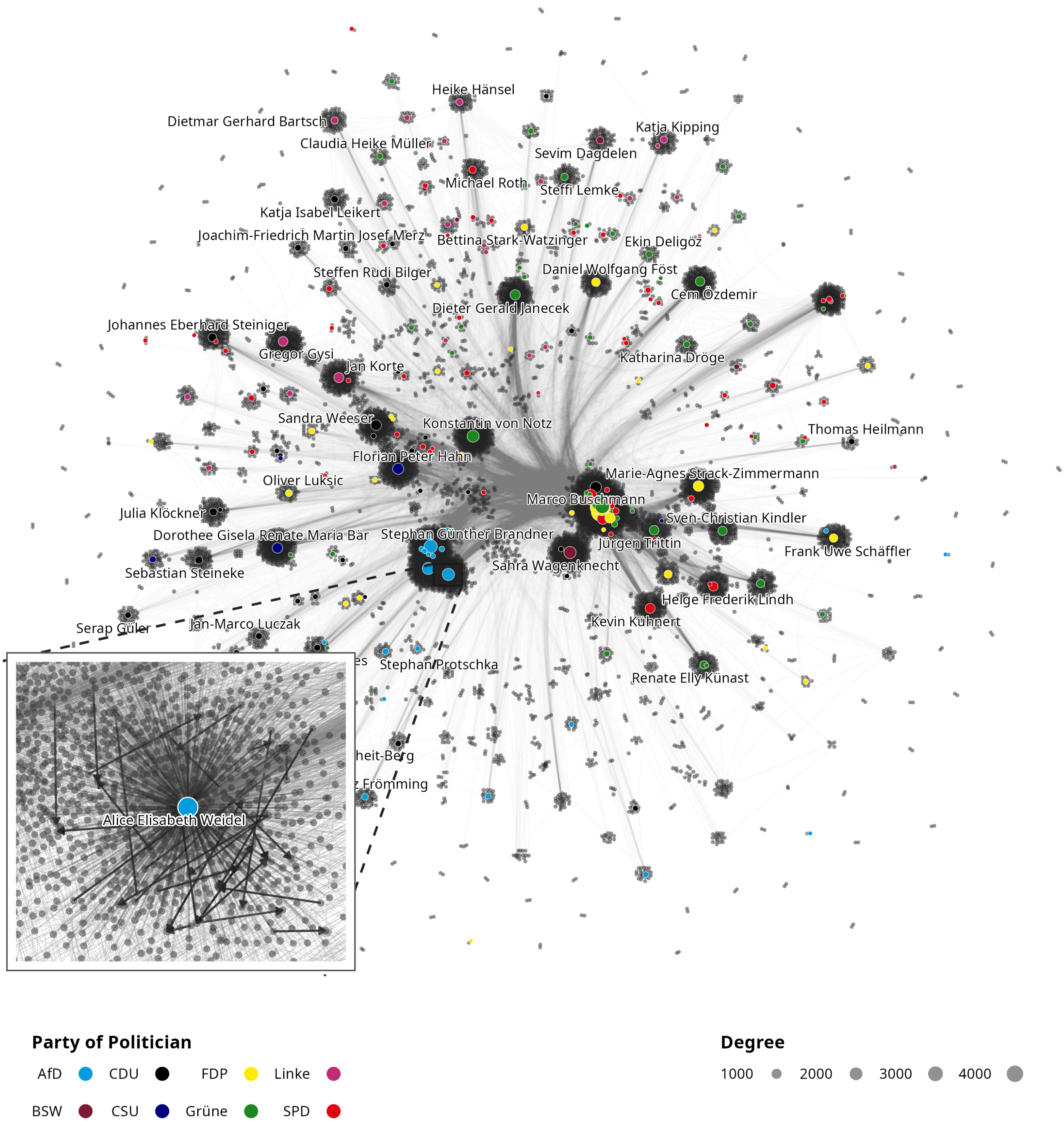

Figure 10: Giant Component of Reply Network

Reply Network laid out with the DrL Algorithm (Martin et al. 2007). Nodesizes scale with degree, politicians are colored by their partycolor. The zoomed inset highlights the directed edges between alters in Alice Weidels' egonetwork.

Figure 10 visualises the giant component of the reply network. The network is laid out with the Distributed Recursive Layout (DrL) algorithm (Martin et al. 2007). At first a force directed network is laid out, which is then iteratively hierarchically clustered into fewer nodes until a sufficiently small network is created. In the end the finer graphs are drawn in again.

Structurally a comparatively small set of high-degree politicians functions as a dense core of the network. Each politician is surrounded by a close reply-community. Some of the communitymembers are then also active in other politicians reply networks connecting them with the core. It almost seems like an ideal hierarcical network in (Ravasz and Barabasi 2002)'s sense, with the difference that this network has very low clustering (Table 2) due to broadcasting structure around each politician.

The hub-periphery structure is divided by a political boundary. Politicians of the parliamentary opposition (Die Linke, BSW, CDU/CSU, AfD) populate the left half of the figure, while members of the governing coalition (SPD, Grüne, FDP) cluster in the dense core and extend toward the right. Politicians whose audiences overlap with those of governing MPs are pulled into the centre, whereas those embedded primarily within partisan audiences are displaced toward the periphery, so that the layout itself recovers a coarse government–opposition distinction.

The inset zooms into Alice Weidel's egonetwork and displays the broadcast pattern at the local level. The overwhelming majority of her alters are connected to her by direct replies to Weidel, but a non-trivial share of edges runs between alters themselves. This within-ego connectivity indicates that beyond pure dyadic broadcasting, smaller communities of repliers also interact with one another, hinting at community activity.

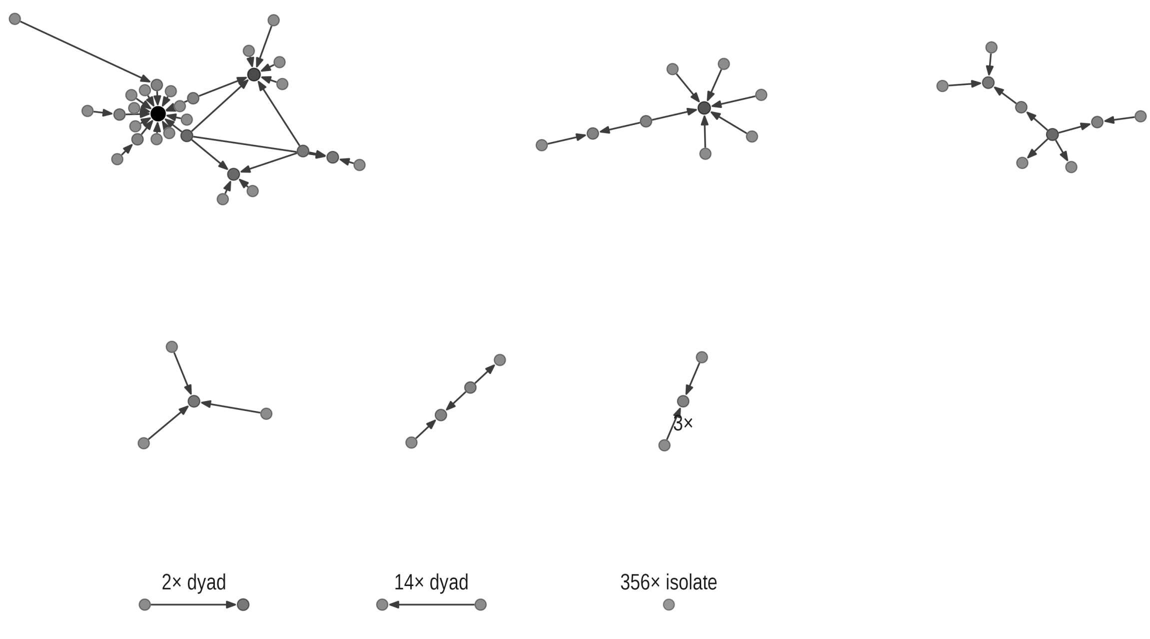

Figure 11: Ego Network - Alice Weidel

Egonetwork up to order 5. All reply-links created beneath other politicians original-tweets removed from a respective politicians egonetwork.

Figure 11 decomposes Alice Weidel's egonetwork up to order 5 once all reply-links produced beneath other politicians original tweets have been pruned, so that only the ties formed within Weidel's own conversational threads remain. The majority of Weidel's engagement-community do not interact with one another within her reply space. At the same time, the connected component and the handful of intermediate structures indicate some interaction patterns among the community. Remember, that all of the alters visualized also have replied to Weidel at some point in the observed week.

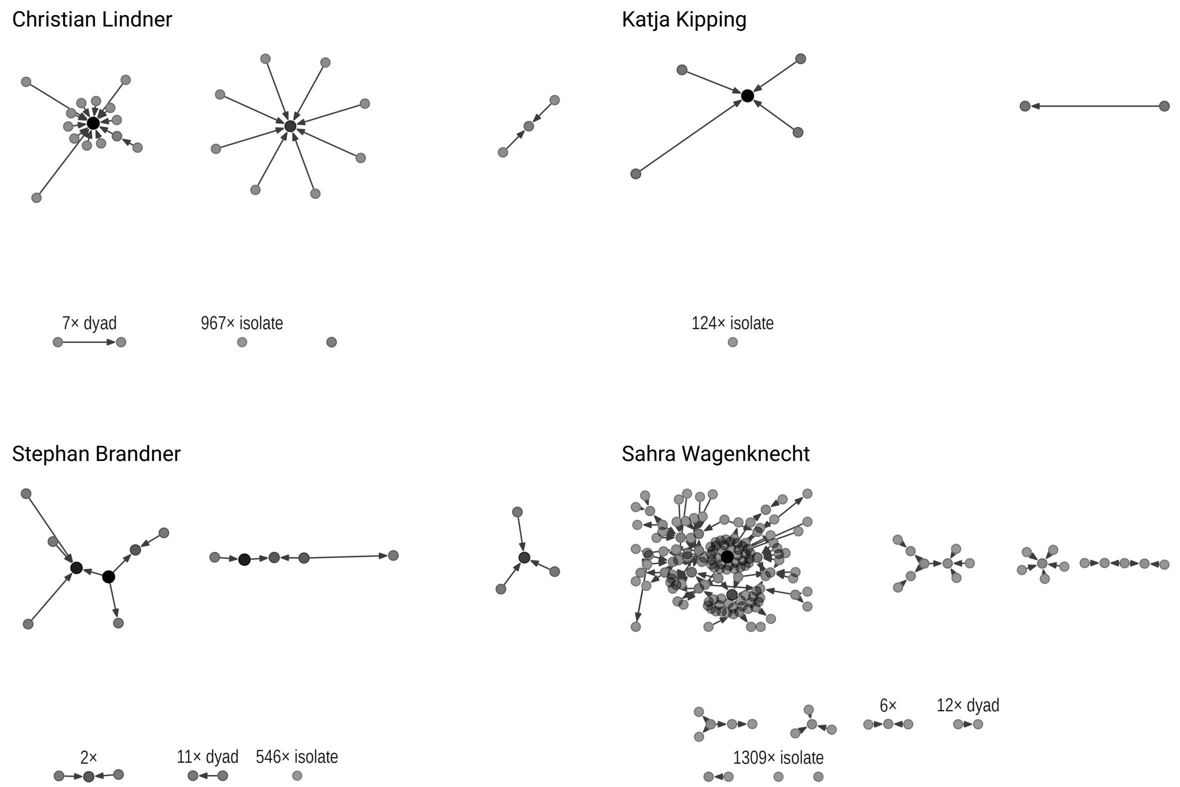

Figure 12: Ego Networks of four Politicians

Egonetworks up to order 5 for Christian Lindner, Katja Kipping, Stephan Brandner and Sahra Wagenknecht. All reply-links created beneath other politicians' original tweets have been removed from each respective politician's egonetwork.

4.3. Egonetwork Analysis

In order to investigate the connection between politicians populist language and the manifestation of the people ingroup in their respective engagement community regression models are estimated.

The models are estimated on the summarized egonetworks of the full network. Of 250 politicians in the retweet network that were part of the 19th, 20th or 21st German Bundestag only 148 started a reply thread in the observed week of 7th to 14th February 2022. Of those 148 only 10 had a populism score above 0.

Score	Mean	SD
Populism	1.03	1.35
People	0.36	0.29
Elite	– 0.50	0.34
Antagonism	0.80	0.68

Party	n
AfD	6
BSW	1
Die Linke	2
SPD	1

Table 3: Descriptive statistics of the 10 populists politicians starting a reply thread.

Since the aim of the analysis is to investigate the impact of populist language on local structures the most feasible way is to binarize the populism score. Thereby the regression models compare the mean connectivity between populist and non-populist networks.

All models regress mean alter degree on the binarized populism score, comparing average alter interconnectedness between populist and non-populist ego networks. Mean alter degree is chosen as a measure, because it is less size-dependent than density which is important since the ego networks size varies quite a lot across politicians. Density is inversely related to network size, making cross-ego comparisons unreliable. As a robustness check, the same specification is re-estimated with the fragmentation ratio as the dependent variable, validating the increased connectivity through another proxy variable capturing how connected alters are among each other.

Three control variables are included. Ego degree captures the politician's structural prominence in the reply network and accounts for the possibility that highly central politicians attract denser alter communities simply by virtue of their position. Follower count proxies platform-level visibility and controls for popularity effects driven by the platforms functionality and the algorithm. Mean thread size of the politician's reply threads controls for the mechanical fact that longer threads provide more opportunities for alter-alter interaction to occur. Assumption checks of the linear regressions are attached in the Appendix.

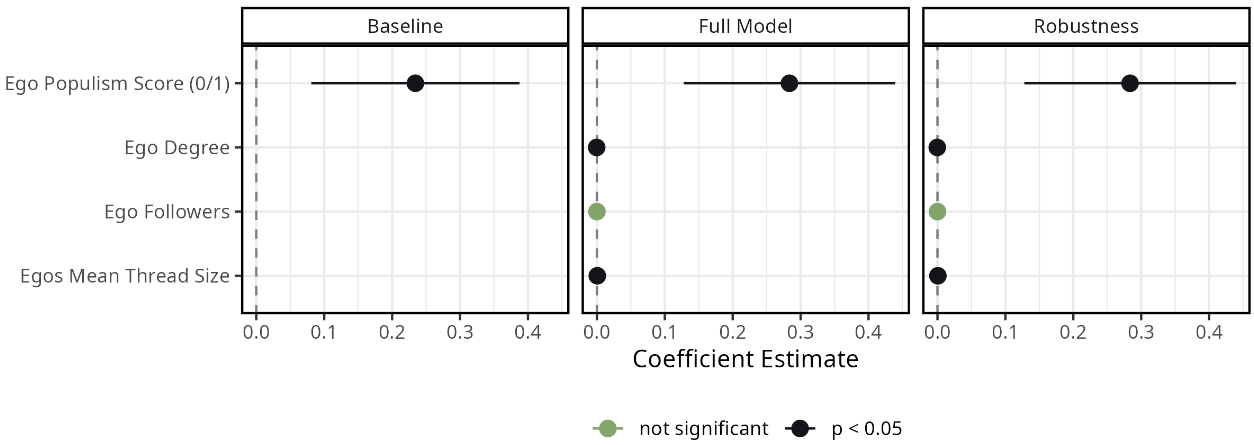

Figure 13: Estimating Mean Alter Degree with Politicians Populist Language

Across all three specifications the binarized populism score emerges as the only substantively meaningful predictor of mean alter degree. The baseline places the effect around

0.25, and the inclusion of the three controls in the full model leaves both magnitude and significance essentially unchanged. Ego degree and mean thread size are statistically significant but with coefficients near zero not contributing explanatory weight. Follower count is indistinguishable from zero throughout. The robustness specification, regressing the fragmentation ratio on the same predictors, reproduces the populism coefficient at comparable magnitude and significance, indicating that the structural signal is not an artefact of the chosen connectivity measure.
The visibly wider confidence interval around the populism estimate, contrasted with the tight bounds on the controls, reflects the small number of populist ego networks in the sample and points toward an effect that is stable in direction across specifications but identified off a limited subset of cases.

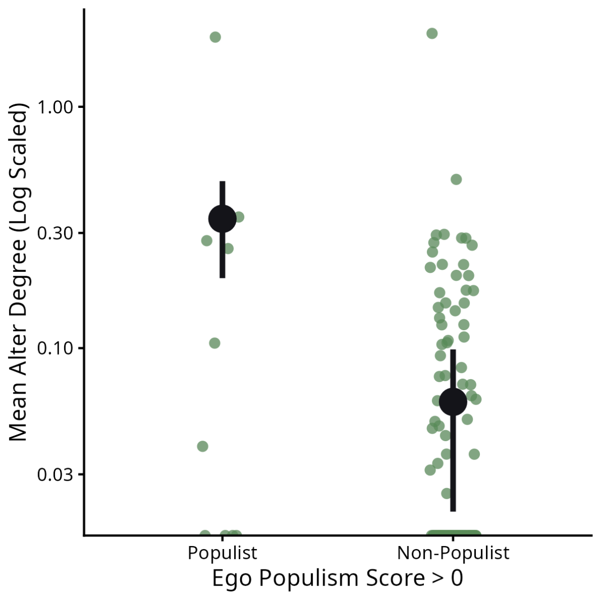

Figure 14: Predicted Values of Mean Alter Degree by politicians (ego) binarized populism score from Full Model. 95% confidence intervals are included in the visualization of the means.

Figure 14 visualizes the predicted means from the full model with the `prediction` function of the marginal effects package (Arel-Bundock et al. 2026). It visualizes the discrepancy of means and samplesizes between the populist and non-populist ego networks. The figure makes the impact of populist vocabulary tangible. Populist rated politicians produce almost three times as connected engagement-communities compared to non populist politicians. The limitations of the small sample size are clearly visible – the few displayed populist egonetworks could very well be an α (Type I) error and have occured by random chance. Additionally to the data being sparse a lot of politicians only get extremely few replies.

While this answers the main hypothesis the next part takes a step further and attempts to explain the "effect" further through investigating the role of alters. The regression models only controlled for ego characteristics. Could there be differences between users engaging with populist content compared to users who engage with non-populist politicians?

To probe this, the alter edge-level analysis contrasts each observed statistic if it is driven by homophily/heterophily (`absdiff`) and activity/prestige (`nodecov`) pairs against a random-pairing baseline within the same ego network. Finally yielding in relative deviations that are comparable across politicians and can be aggregated by group.

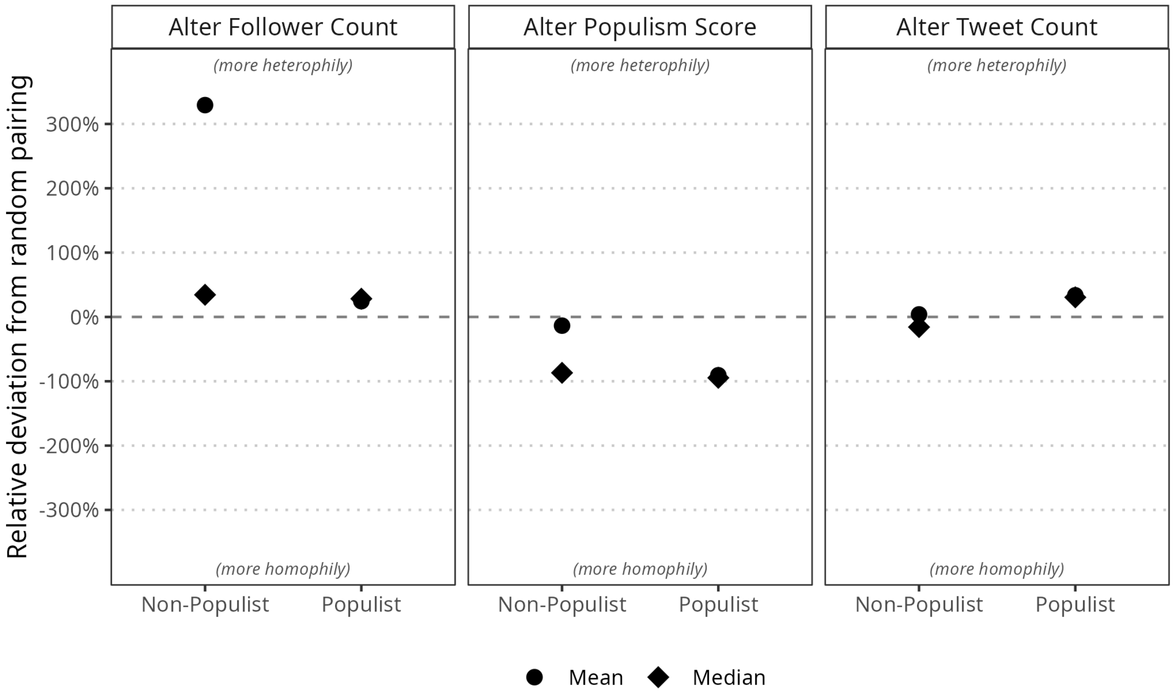

Figure 15: Edge-level alter homophily relative to random pairs of the same alters, separated by ego type.

Figure 15 shows the absolute differences between alters on a given statistic. This allows for the exploration of homophilic/heterophilic interactions in the alternetworks.

On the follower count, both ego types display heterophily. The absolute difference between connected alters lies roughly 25-35% above random at the median for both populist (+28%) and non-populist (+34%) egos, while the non-populist mean (+329%) is heavily inflated by a few outlier networks. Connected alters differ ~30% more than random pairs in follower count across both ego types, with a small set of non-populist networks pushing the mean far above this. Users who reply to each other thus tend to span different audience sizes, large and small accounts mix on edges rather than clustering with their own kind.

On the populism score, both ego types display strong homophily. The absolute difference between connected alters falls ~90% below random in populist ego networks (median and mean) and at the median for non-populists, though the non-populist mean (-14%) is pulled toward zero by a few outliers. Connected alters' populism scores differ 90% less than random pairs in both populist and non-populist ego networks, though a few non-populist outliers pull their mean to −14%. In both populist and non-populist politicians' ego networks, users who reply to each other almost always share similar populism levels. Cross-populism exchanges between alters are rare.

On the tweet count, the patterns diverge by ego type. Populist ego networks show heterophily of 30% above random at both median and mean, while non-populist ego networks sit close to baseline, slight homophily at the median (-16%) and essentially zero at the mean (+4%). Connected alters differ 30% more than random pairs in tweet count in populist ego networks, but only at roughly random levels in non-populist ones. In populist egonetworks, replies tend to bridge users with very different posting volumes, while in non-populist threads users mix across activity levels about as much as chance would predict.

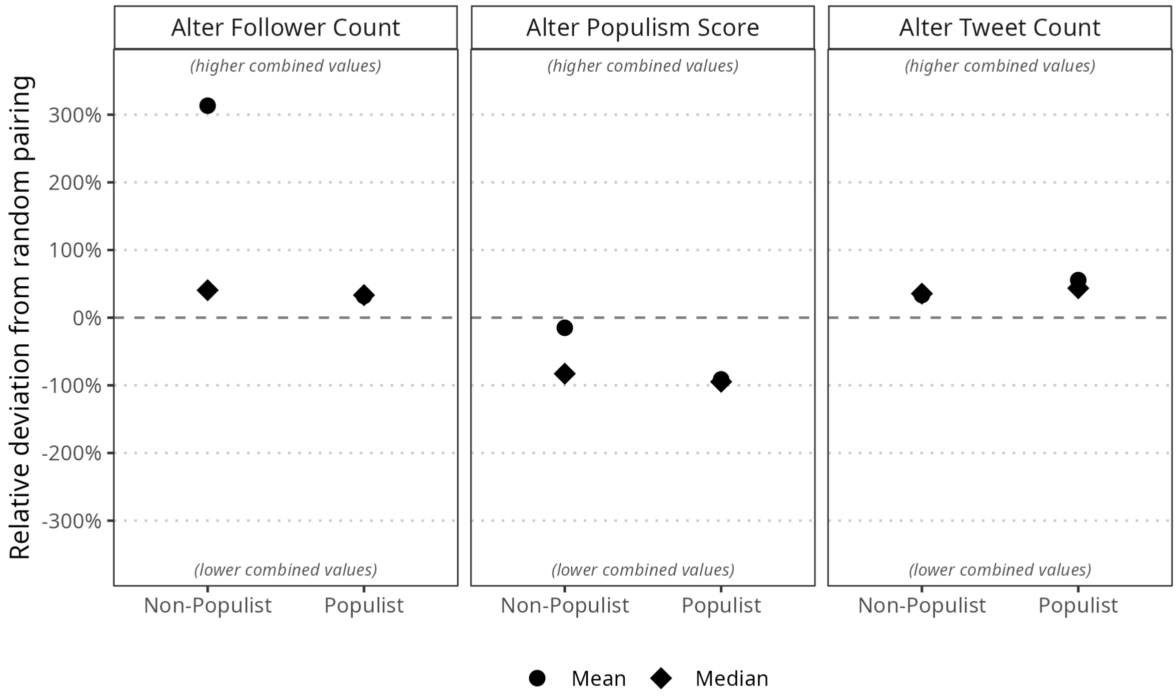

Figure 16: Edge-level alter activity relative to random pairs of the same alters, separated by ego type.

Figure 16 shows the sum of attribute values of two connected alters. This allows for the exploration of activity or prestige effects in the alternetworks.

On the combined follower count, both ego types show positive deviations. Populist ego networks lie ~33% above random for both median and mean, while non-populist egos reach +41% at the median and +313% at the mean, again driven by a few extreme cases. Connected alters' combined follower counts are 30-40% higher than random pairs in both ego types, with non-populist outliers inflating the mean. Replies thus tend to involve at least one well-followed account, engagement clusters around higher-visibility users rather than at the periphery.

On the combined populism score, both ego types show strongly negative deviations. Populist ego networks fall ~95% (median) and ~91% (mean) below random, while non-populist egos reach −83% at the median but only −15% at the mean. Combined populism scores on connected edges are 90% lower than random pairs in populist ego networks and similarly low at the median for non-populist ones, with outliers again pulling the non-populist mean toward zero. Reply pairs thus both sit at low populism. The homophily observed in Figure 15 plays out specifically at the non-populist end of the scale, not through mutual high-populism interaction.

On the combined tweet count, both ego types show positive deviations. Populist ego networks lie ~44% (median) to ~56% (mean) above random, non-populist ones around +34% at both. Connected alters' combined tweet counts are 30-55% higher than random pairs in both ego types, more pronounced in populist networks. Replies thus cluster around the more active users in each thread. High-volume posters draw disproportionately many of the connections.

Across both populist and non-populist politicians, the alters who reply to each other share three traits at the edge level. First, they sit at similar points on the populism scale, overwhelmingly at the low end, so cross-populism exchanges between alters are rare in either ego type. Second, replies bridge rather than match audience sizes. Connected alters differ in follower count more than random pairs would. Third, combined follower and tweet counts on connected edges lie well above random, meaning edges carry more visibility and posting volume than chance would predict. Conversations thus happen mostly between users who agree on populism, but bring together accounts that differ a lot in size and activity, with the more visible and frequently posting users taking part in a disproportionate share of replies. The two ego types diverge only on alter tweet count. Populist networks show heterophily, with replies bridging users of different posting volumes, while non-populist networks sit close to random on this dimension. In threads around populist politicians, replies tend to connect heavy posters with more occasional ones, while in non-populist threads users of all posting volumes mix about as much as chance would predict.

This paints a picture of reply networks around politicians as ideologically narrow but socially mixed spaces, where users overwhelmingly engage with others who share their populism stance but where the conversation is anchored by a smaller set of high-visibility, high-activity accounts pulling in a wider range of audience sizes - a pattern that holds uniformly around populist politicians and, with the exception of tweet-volume mixing, around non-populist ones as well.

5. Discussion

The discussion first summarizes the results and continues with a short summary of the literature review. It ends with interpreting the results through the literature review and in the end adresses the most relevant limitations.

The used data covers reply threads to German MPs' tweets from 7–14 February 2022, a week shaped by the Omicron wave and the war between Russia and Ukraine. Replies form a directed network linking users by their interactions. LLM-based scoring (Figure 4) shows populism is mostly driven by anti-elitism and antagonism and less by pro-people attitudes. Most users do not score on the elite dimension, but those who do are almost all anti-elite and often also pro-people (Figure 5). Party scores reproduce the government–opposition split, with AfD, BSW and Linke most populist and CDU/CSU, FDP, SPD and Grüne least, while outlier politicians in CDU, FDP and AfD pull far above their party mean (Figure 6). Actor vocabulary is shared across populist subcategories but topics differ (Figure 7). `volk` correlates most strongly with the populism score (>0.20), followed by `bürger`, `bevölkerung` and `politiker`.

The reply network is sparse, hub-driven and exhibits a giant component (Table 2). A small set of high-degree politicians absorbs replies from a vast low-degree periphery, edges flow almost entirely one way, and high-degree nodes connect to low-degree ones rather than to each other. The network is sparse and local cohesion is essentially absent. A modular core–periphery structure sits on top, partitioning cleanly into well-separated communities (Figure 8, Figure 9). While the giant component is globally divided by a government–opposition structure the core holds politicians of government parties like Marco Buschmann (Figure 10, Figure 12).

Of the 148 MPs active in this window, only ten score as populist, drawn from AfD, BSW, Linke and SPD (Table 1). OLS regressions of ego networks on mean alter degree suggest higher populism scores are linked to denser alter–alter connectivity, though small N limits the result. Alters effect on reply-edge formation show connected alter pairs generally share populism scores but bridge differences in account size, visibility and reach, indicating more active accounts pull the periphery into engagement. The one populist/non-populist divergence lies in tweet volume, where populist engagement communities connect users of differing activity levels more than non-populist ones do.

Populism in this thesis is understood through a minimal definition as the moral divide between the good, powerless people that stand against the corrupt, powerful elite (Mudde 2004). Previous research already found social media as a low affordance alternative media for populists to broadcast their ideas (Gil De Zúñiga et al. 2020). This usage of alternative media has become part of populist group identity (Stier et al. 2025). Politicians use social media to campaign rather than discuss policies, probably because populist frames performed well compared to other strategies in terms of engagement metrics (Cassell 2021; Stier et al. 2018). It has been shown that Dutch politicians usually broadcast their information rather than interact with their community (Jacobs and Spierings 2019). Their audience selfselects themselves in or out of the conversation based on their own evaluations of the situation (Luhmann 1987; Oswald et al. 2025; Tilly 1978; White 2008). In order to form a cohesive group (1) a homogenization process is needed that draws people in and (2) a repulsive process is needed to define clear group boundaries (Stadtfeld et al. 2020). Extending the purely singular engagementmetric based findings this thesis formed an expectation on the meso level:

German MPs who use populist rhetoric have higher interconnected alters in their reply ego networks compared to MPs using less populist rhetoric.

The results of the ego network analysis support to this expectation. OLS regressions of mean alter degree show a positive association with populism scores, indicating that populist MPs do sit in more interconnected reply communities than less populist ones. The alter effects on reply formation in the ego network add texture to the result. Connected alter pairs generally share populism scores but bridge differences in account size, visibility and reach, and the one feature on which populist and non-populist engagement communities diverge is tweet volume, where populist communities connect users of differing activity levels more than non-populist ones do. Read through the dual logic of group formation (Stadtfeld et al. 2020), this pattern looks like homogenization on the dimension that defines the in-group, namely the populism frame itself, paired with a repulsive boundary that distinguishes populist from non-populist communities while leaving size and activity heterogeneous on the inside. The populist frame thus appears to be what allows a low-activity periphery to attach itself to a high-activity core in the first place, knitting otherwise unequal users together around a shared antagonism.

At the macro level the network displays the broadcasting logic that previous research has attributed to political communication on social media (Jacobs and Spierings 2019). A sparse, hub-driven topology with negligible reciprocity or local clustering is exactly what one expects when politicians use the platform to campaign rather than deliberate (Stier et al. 2018). The clean partition into well-separated communities along the government and opposition divide fits self-selection into closed observation contexts. Populist rhetoric concentrating in AfD, BSW and Linke, with outliers in CDU, FDP and AfD far above their party mean, matches the finding that populist framings have become part of a group identity. Beyond the selfselection this global network structure as well as textual framing captures the moral loading of the representative democratic systems structures of the german parliament. Taken together, the macro broadcasting logic, the meso ego network density, and the lexical centrality of antagonism point to social media functioning as the low affordance alternative arena described in earlier work (Gil De Zúñiga et al. 2020), where populist accounts not only broadcast louder but also organise a denser and more cohesive audience structure around themselves.

The analysis comes with several limitations. First of all the dataset only spanned one week giving room to external events impacting the discourse on the platform, possibly distorting the signal of the text analysis. Additionally there were only a small number of populist posting MPs captured during this week. This limits the interpretability of the ego network analysis drastically and reduces the finding to a pointer. Future work could replicate Oswald et al. (2025)s experimental result in a real-world setting estimating a similar hurdle model. The text analysis is limited by the aggregation of tweets on the actor level. It could be that politicians anti-elitist tweets are about unrelated topics in comparison to pro-people tweets. Still for politicians with a high share of populist tweets the presumption of a political communicative strategy gets stronger. As with all social media studies no inference to the analog world is sufficiently possible because of the impact of the algorithm as well as the selfselection of users being active on the platform.

6. Conclusion

The analysis revealed that politicians using populist rhetoric on twitter are surrounded by a stronger connected engagement community compared to non-populist politicians. Users forming replies in populist communities do largely not differ from users of non-populist communities in terms of mean behaviour. While populist rhetoric is morally charged and often antidemocratic, the populist landscape on social media is shaped by existing government–opposition structures.

Three implications follow from the conducted analysis. First, populism online is not just a louder broadcast but a logic of organisation, since the ego network connectivity result shows the populist frame builds audience structure on the meso level and not only stay at the engagement-metric level. Second, politicians provide opportunity for homogenisation around them based on topics and rhetoric. It seems that shared antagonism rather than shared platform behaviour binds populist communities. Third, social-media platforms reproduce parliamentary structure rather than dissolve it. This mapping is driven by individual politicians that are found more often in the usually as "populist" described parties than in government parties, even though also these have populist posting actors.

The results emphasize the advantages of studying populism as a type of communication connecting observational studied with classical sociology exploring the social with a focus on the meso scale. Populist rhetoric on social media seems to be a successful tool for gathering online engagement communities, having potential to improve or threaten democratic structures.

Appendix A: Additional Tables, Figures and Equations

symmetry of F1 Score for swapping raters

Given binary ratings $a_i, b_i \in \{0, 1\}$ from two raters, define:

$$\text{both\_positive} = \sum_i a_i b_i$$

$$\text{only}_{b_{\text{positive}}} = \sum_i (1 - a_i) b_i$$

$$\text{only}_{a_{\text{positive}}} = \sum_i a_i (1 - b_i)$$

Treating A as ground truth:

$$\text{precision} = \frac{\text{both\_positive}}{\text{both\_positive} + \text{only}_{b_{\text{positive}}}}, \quad \text{recall} = \frac{\text{both\_positive}}{\text{both\_positive} + \text{only}_{a_{\text{positive}}}}$$

Swapping roles swaps $\text{only}_{a_{\text{positive}}} \leftrightarrow \text{only}_{b_{\text{positive}}}$, which swaps precision and recall. Since F_1 is their harmonic mean:

$F_1 = \frac{2\cdot\ \text{precision}\ \cdot\ \text{recall}}{\text{precision}\ +\ \text{recall}} = \frac{2\cdot\ \text{recall}\ \cdot\ \text{precision}}{\text{recall}\ +\ \text{precision}}$

Multiplication and addition are commutative, so F_1 is invariant under the swap. ∎

For swapping labels (0,1) F Scores are not symmetric.

Component Tables

Component Size	Number of Components	Total Nodes	% of Network
77194	1	77194	94.96%
41	1	41	0.05%
34	1	34	0.04%
31	1	31	0.04%
27	1	27	0.03%
23	1	23	0.03%
19	1	19	0.02%
16	1	16	0.02%
14	1	14	0.02%
11	3	33	0.04%
10	2	20	0.02%
9	4	36	0.04%
8	6	48	0.06%
7	7	49	0.06%
6	13	78	0.10%
5	24	120	0.15%
4	46	184	0.23%
3	185	555	0.68%
2	1207	2414	2.97%
1	359	359	0.44%

Table 4: Component size distribution of the reply network

linking users through directed and weighted reply links. A giant component holds most user nodes and the size appearing the most is two.

Textanalysis Additional Results

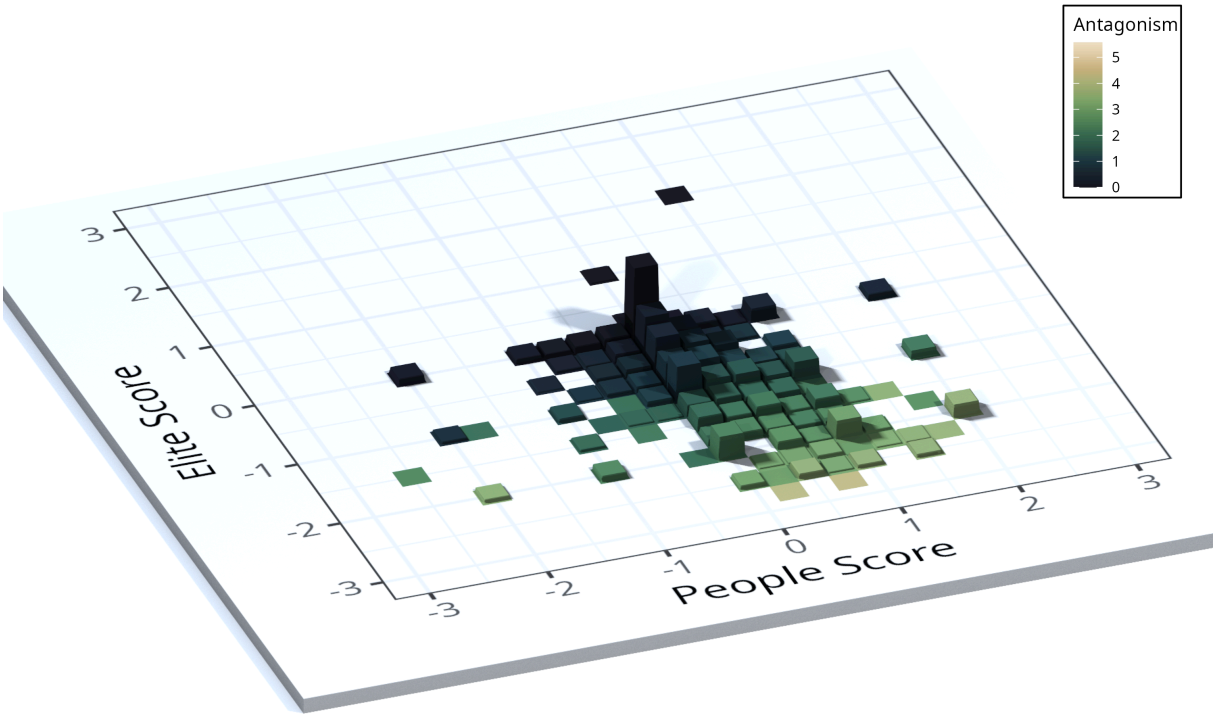

Figure 17: Distribution of populism dimensions per user in the giant component;

People score ($\bar{p}_u$) and elite score ($\bar{e}_u$) are binned into 0.25-wide intervals. Color visualizes mean antagonism ($\bar{a}_u$) per bin; height describes user count ($n = 29\,672$).

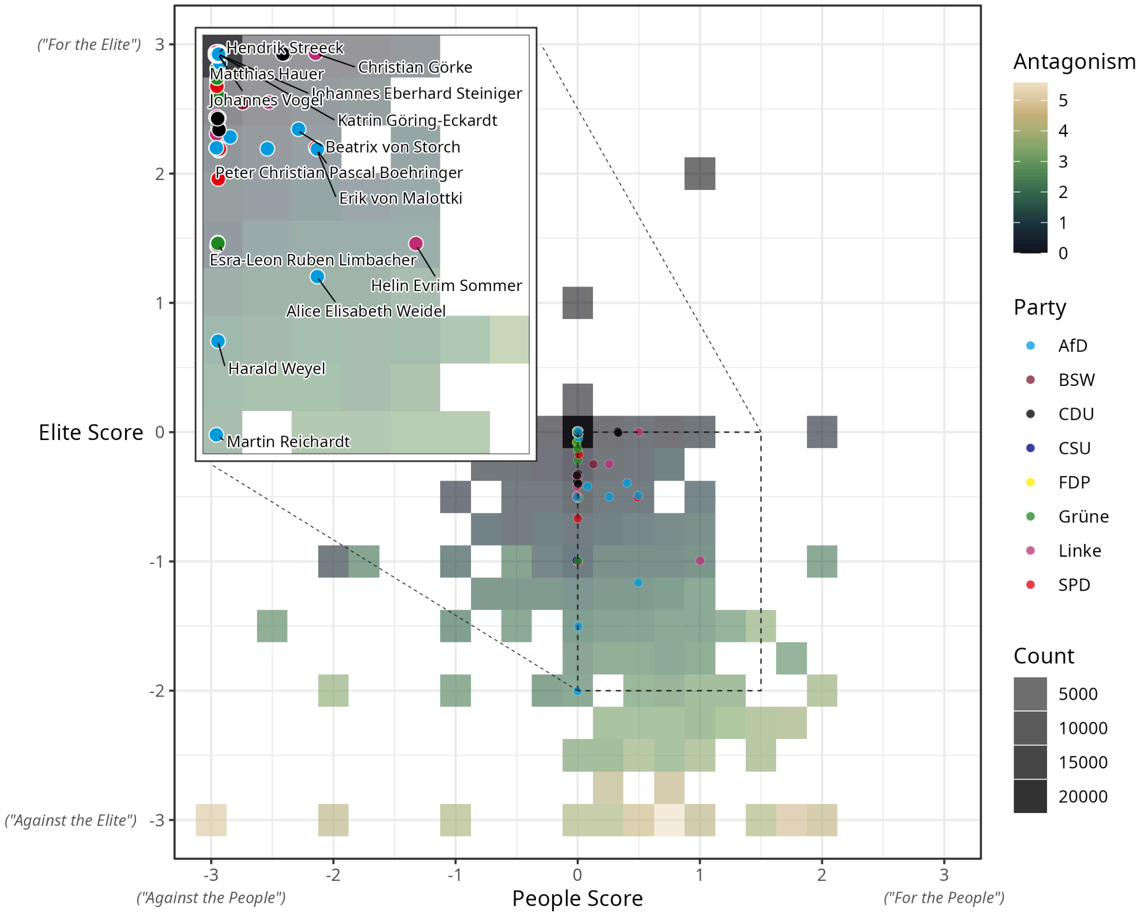

Figure 18: Distribution of populism dimensions per user in the giant component with marked politicians position;

People score ($\bar{p}_u$) and elite score ($\bar{e}_u$) are binned into 0.25-wide intervals. Color visualizes mean antagonism ($\bar{a}_u$) per bin; opacity describes user count ($n = 29\,672$).

Regression Model checks

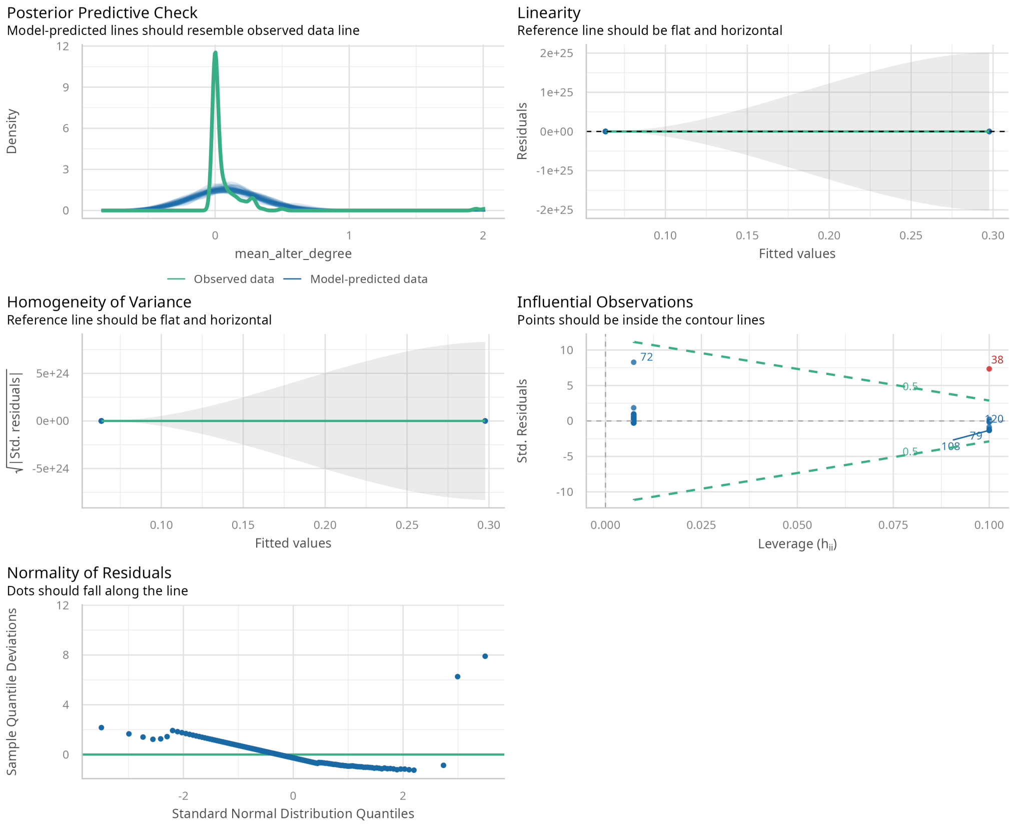

Figure 19: Assumption Check - M0 (Baseline)

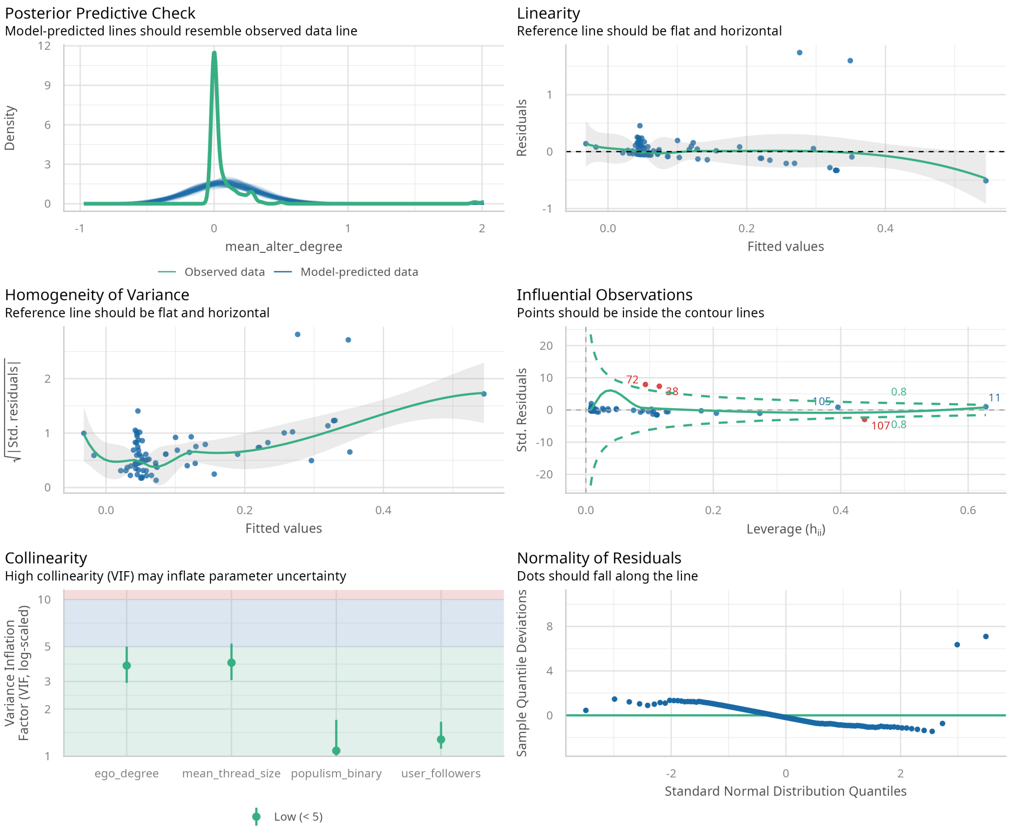

Figure 20: Assumption Check - M1 (Full Model)

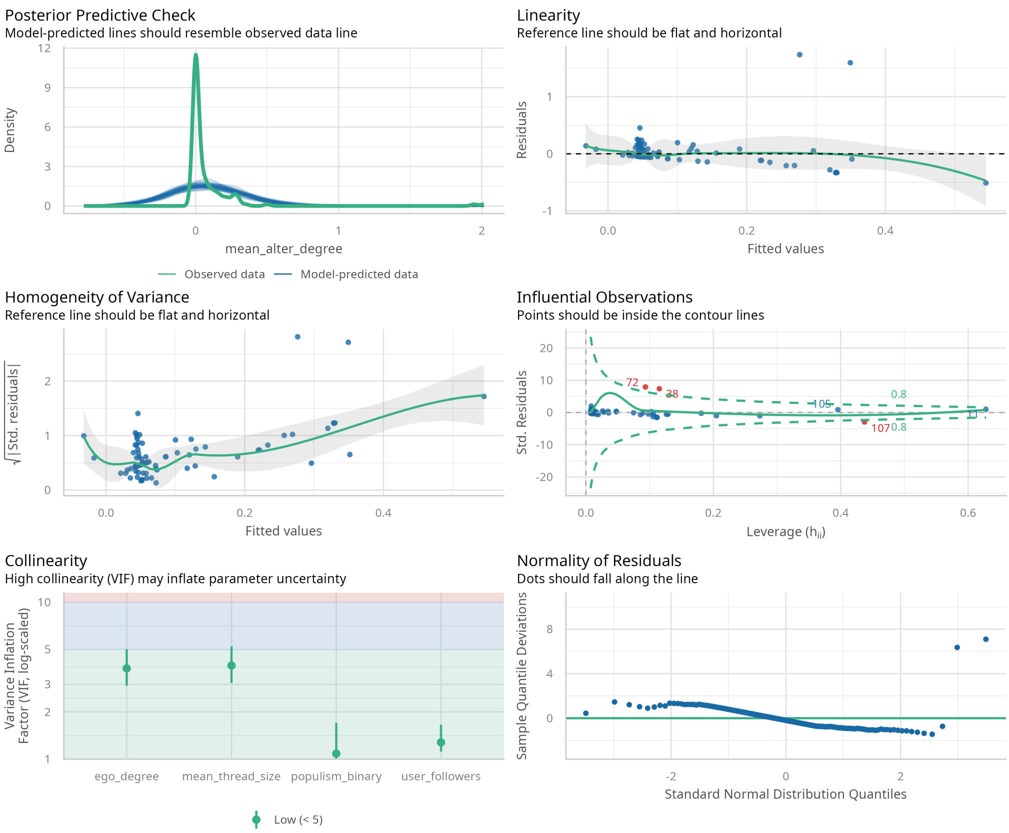

Figure 21: Assumption Check - R1 (Robustness)

Appendix C: Prompt

```
You are a political sociology analyst trained to detect populist rhetorical strategies in social media posts. Your
goal is to identify which posts contain dimensions of populist attitudes — without judging political views.

Analyze each post holistically. Avoid speculation.

---
### Pre-Analysis Check
Before analyzing, ask: **Does this text carry a detectable rhetorical signal** — through language, symbols, or clear
implication?

If you find no signal at all (e.g., only user mentions, hashtags, URLs, or emojis with no interpretable stance),
return the JSON with all scores as 0 and the holistic redescription as: "No semantic argument detected."

If the signal is ambiguous but present, proceed with analysis and flag uncertainty.

---

### Dimensions of Populism

1. **People Attitude (The Ingroup)**
    - **Definition:** Arguing in favour (positive attitude) or against (negative attitude) a large homogenous ingroup
("The People") considered the societal norm.
    - "The People" must imply the **universal ordinary majority** — e.g., "The common people," "The citizens," "We,"
"Us," "This country's people."
    - A single individual or a list of named individuals is not "The People."
    - Specific subgroups (e.g., "families," "pensioners," "minorities") are not "The People" on their own.

2. **Elitist Attitude (The Outgroup)**
    - **Definition:** Arguing in favour (positive attitude) or against (negative attitude) a small, powerful
outgroup.
    - "The Elite" refers to a powerful class or group — e.g., "politicians," "the ruling class," "the establishment,"
"the media."

3. **Antagonism (The Divide)**
    - **Definition:** Antagonism captures the strength of opposition between The People and The Elite.

---

### Analysis Instructions

#### Step 1: Holistic Redescription** (concise — aim for roughly one paragraph)
- Synthesize the post's strategic purpose:
  1. Summarize the core narrative.
  2. Identify the rhetorical structure.


#### Step 2: Actor & Signal Analysis
Identify every person or group referenced, then work through these questions for each:

1. **Scale:** Is this actor an individual, a named role or institution, or a generalized class? _How confident am I —
what would the alternative reading be?_

2. **People signal:** Does the text invoke a broad ordinary majority? _Or am I inferring "The People" just because an
elite is attacked? If the text names a specific subgroup rather than "The People" as a whole: does the text frame
this subgroup as representing the ordinary majority's experience, or is it advocating for that group's particular
interests?_

3. **Elite signal:** Does the text target a powerful group or class? _Or is it criticizing a specific policy or a
single person without generalizing?_

4. **Antagonism signal:** Does the text construct opposition between people and elite? _How strong is the moral
charge — is the elite merely wrong, or morally culpable, or an existential threat?_

If a reasonable coder could disagree with your classification on any of these, state the alternative interpretation
and flag your confidence as LOW. Otherwise, flag HIGH.


#### Step 3: Scoring
Score each dimension using the scales below.

**People Attitude (The Ingroup)**
```

````
- **+3:** "The People" are viewed as virtuous or superior.
- **+2:** Clear support for "The People."
- **+1:** Mild support.
- **0:** Neutral or absent.
- **-1:** Mild criticism of "The People."
- **-2:** Clear criticism of "The People."
- **-3:** "The People" are viewed as incompetent or irrational.

**Elitist Attitude (The Outgroup)**
- **+3:** "The Elite" is viewed as virtuous or superior.
- **+2:** Clear support for "The Elite."
- **+1:** Mild support.
- **0:** Neutral or absent.
- **-1:** Mild criticism of "The Elite."
- **-2:** Clear criticism of "The Elite."
- **-3:** "The Elite" are viewed as incompetent or irrational.

**Antagonism (The Divide)**
- **0 (No divide):** No opposition is framed between the people and the elite.
- **1–2 (Dissatisfaction):** Passive complaints or implied asymmetry without a moral charge.
- **3–4 (Active blame):** The elite is framed as responsible and culpable; characterized by insults, accusations of
dishonesty, and explicit us-vs-them framing.
- **5–6 (Existential threat):** The elite is framed as an enemy; includes dehumanization, conspiratorial framing, and
calls for collective action or eliminationist language.



### Few-Shot Examples: Populist Rhetoric Analysis

**Example 1: Strong Populist Signal**

_INPUT TEXT TO ANALYZE:_
> "The hardworking people of this country are sick and tired of corrupt politicians lining their pockets while
ordinary families can't afford groceries. It's time WE take our country back from the Washington elites who have
rigged the system against us. They don't care about you — they never did."

_OUTPUT:_
```json
{
  "holistic_redescription": "The post constructs a classic populist narrative pitting a virtuous, suffering ordinary
majority ('the hardworking people,' 'ordinary families,' 'WE,' 'us,' 'you') against a morally corrupt political elite
('corrupt politicians,' 'Washington elites'). The rhetorical structure frames the elite as deliberately exploitative
('lining their pockets,' 'rigged the system') and emotionally indifferent ('they don't care about you'), while
issuing a mobilizing call to collective action ('take our country back'). The moral charge is strong: the elite is
not merely incompetent but actively culpable.",
  "actor_analysis": "1. 'The hardworking people of this country' / 'ordinary families' / 'WE' / 'us' / 'you' —
generalized class representing the broad ordinary majority. Confidence: HIGH. 2. 'Corrupt politicians' / 'Washington
elites' — generalized class representing a powerful political outgroup. Confidence: HIGH. No individual is named; the
criticism targets the political class as a whole.",
  "people_explanation": "A broad ordinary majority is explicitly and repeatedly invoked through multiple framings:
'hardworking people of this country,' 'ordinary families,' and inclusive pronouns ('WE,' 'us,' 'you'). These are not
specific subgroups advocating particular interests — they represent the universal ordinary citizen. The People are
framed as virtuous ('hardworking') and suffering ('can't afford groceries').",
  "people_score": 2,
  "elite_explanation": "A powerful political class is explicitly targeted as a generalized outgroup — 'corrupt
politicians' and 'Washington elites.' The criticism is not directed at a single individual or a specific policy but
at the political establishment as a whole. They are characterized as morally corrupt ('lining their pockets'),
deliberately manipulative ('rigged the system'), and callously indifferent ('don't care about you').",
  "elite_score": -3,
  "antagonism_explanation": "The text constructs sharp, moralized opposition between The People and The Elite. The
elite is framed as actively culpable — they have 'rigged the system' and are 'corrupt.' The language escalates beyond
dissatisfaction into active blame with a mobilizing call ('take our country back'), suggesting the elite's control is
near-illegitimate. The moral charge is strong (culpable and indifferent), though the post stops short of
dehumanization or conspiratorial framing, placing it in the upper range of active blame.",
  "antagonism_score": 4
}
```

---

**Example 2: No Detectable Populist Signal**

_INPUT TEXT TO ANALYZE:_
> "Just finished watching the new season of that baking show 🎂 Honestly episode 4 was wild. Anyone else think Marco
should have stayed longer? #BakeOff #TeamMarco"

_OUTPUT:_
````

````
```json
{
  "holistic_redescription": "No semantic argument detected.",
  "actor_analysis": "One individual is mentioned ('Marco'), who is a contestant on a television show. No political
actors, institutions, generalized classes, or power dynamics are referenced. Confidence: HIGH.",
  "people_explanation": "No broad ordinary majority is invoked. The post is a personal entertainment opinion with no
political or societal framing.",
  "people_score": 0,
  "elite_explanation": "No powerful group or class is referenced or targeted. The post contains no political or
institutional criticism.",
  "elite_score": 0,
  "antagonism_explanation": "No opposition between any groups is constructed. The post is entirely about a television
program.",
  "antagonism_score": 0
}
```

---

**Example 3: Ambiguous / Mild Signal**

_INPUT TEXT TO ANALYZE:_
> "Another satisfying day of the government announcing new taxes while somehow never fixing the roads or the schools.
But sure, keep telling us everything is fine. We're not blind."

_OUTPUT:_
```json
{
  "holistic_redescription": "The post expresses frustration with government performance, using sarcasm ('But sure,
keep telling us everything is fine') to highlight a perceived gap between government claims and lived reality. The
rhetorical structure implies a contrast between an underserved public ('us,' 'we') and an ineffective government, but
the framing remains at the level of policy dissatisfaction – criticizing specific failures (roads, schools, taxes) –
rather than constructing a full populist morality tale. The 'us' vs. government framing is present but mild.",
  "actor_analysis": "1. 'The government' – a named institution, not a generalized elite class. However, 'the
government' could be read as standing in for the broader political establishment. Confidence: LOW – a reasonable
coder could interpret this as either targeted institutional criticism or broader anti-elite sentiment. 2. 'Us' / 'We'
– an inclusive pronoun suggesting the general public. Could represent the broad ordinary majority or simply the
speaker and their community. Confidence: LOW – the referent is ambiguous between a universal 'people' and a more
local or personal 'us.'",
  "people_explanation": "The pronouns 'us' and 'we' suggest a broad public, and 'We're not blind' implies the
ordinary citizen sees through official narratives. However, the post does not explicitly frame 'The People' as a
virtuous or unified majority – it is closer to a shared complaint from a general public perspective. The signal is
present but mild, and a coder could reasonably score this as 0 (just a personal complaint) or +1 (mild invocation of
ordinary people's experience).",
  "people_score": 1,
  "elite_explanation": "The criticism targets 'the government' as an institution, focusing on specific policy
failures (taxes, roads, schools) and dishonesty ('telling us everything is fine'). This sits between criticizing a
specific institution's performance and painting a broader elite as out of touch. Because 'the government' is a single
institution and the complaints are policy-specific, this leans toward institutional critique rather than generalized
anti-elite framing. Score reflects mild criticism of a powerful institution without full generalization to an elite
class.",
  "elite_score": -1,
  "antagonism_explanation": "There is an implied asymmetry: the government collects taxes but fails to deliver, and
it dismisses the public's awareness ('telling us everything is fine' vs. 'we're not blind'). This constructs a mild
divide – the government is framed as dismissive and ineffective, and the public is positioned as the aware,
underserved party. However, the moral charge is limited: the government is incompetent and possibly dishonest, but
not painted as existentially threatening or conspiratorially corrupt. This fits the dissatisfaction range.",
  "antagonism_score": 2
}
```

# Output Format

Only answer with a codeblock containing a JSON file in the following format:
```json
{
  "holistic_redescription": "Synthesize the post's core narrative and rhetorical structure (roughly one paragraph).",
  "actor_analysis": "List all actors. For each: individual, role/institution, or generalized class? State confidence
(HIGH/LOW) and note alternative readings where applicable.",
  "people_explanation": "Is a broad ordinary majority invoked – or are you inferring 'The People' from other signals?
If a subgroup is named, does it stand in for the majority or represent particular interests?",
  "people_score": 0,
  "elite_explanation": "Is a powerful group or class targeted – or is the text criticizing a specific policy or
individual without generalizing?",
  "elite_score": 0,
  "antagonism_explanation": "Does the text construct opposition between people and elite? How strong is the moral
charge – merely wrong, culpable, or existential threat?",
  "antagonism_score": 0
````
````

~~~
}
```
~~~

INPUT TEXT TO ANALYZE:

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